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Illustrated Textbook of Paediatrics

Fourth Edition



Tom Lissauer
Graham Clayden

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Fourth Edition



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Illustrated Textbook of Paediatrics

Fourth Edition

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Foreword

When the late Frank A. Oski wrote the foreword for the first edition of this book in 1997, he gave it fulsome praise and predicted that it would become a 'standard by which all other medical textbooks will be judged'. He was a great man and a wonderful writer, so his prediction was no doubt welcomed by the authors, both well known for their contribution to undergraduate and postgraduate medical education and assessment.

I have a much easier task in writing the foreword for the fourth edition in 2011. The mere fact that there is a fourth edition is testimony in itself, but there is also the fact that this book has become the recommended paediatric textbook in countless medical schools throughout the world and has been translated into 8 languages. I have travelled the world over the last 15 years and wherever I have been in a paediatric department, the distinctive sunflower cover of 'Lissauer and Clayden' has been there with me. Whether it is Hong Kong, Malaysia, Oman or South Shields, it is there!

It is not surprising that it has won major awards for innovation and excellence at the British Medical Association and Royal Society of Medicine book awards. The book is well established and widely read for the simple reason that it is an excellent book. Medicine is now so complex and information so vast that students are no longer expected to know all there is to know about medicine. What they need are the core principles and guidance as to where to find out more. This book gives the core principles, but also provides a great deal more for the student who wishes to extend his or her knowledge. It is in a very accessible form and has a style and layout which facilitates learning. There are many diagrams, illustrations and case histories to bring the subject to life and to impart important messages. This new edition includes summaries to help revision and also provides access to online assessment tools.

It has been thoroughly updated with many new authors, each of whom is an expert in their own field, but who has been chosen because of their ability to impart the key principles in a non-specialist way. The text focuses on the key areas of paediatrics and new sections include child protection and global health.

There are now countless doctors throughout the world for whom this textbook has been their introduction to the fascinating and rewarding world of paediatrics.

For students, it is all they need to know and a bit more. For postgraduates, it provides the majority of information needed to get through postgraduate examinations and stimulates and guides the reader into the world of clinical paediatrics where practical experience can be gained, built on the sound foundation of the Lissauer and Clayden knowledge base.

The authors are to be congratulated on the continuing success of this book.

I can only echo what Frank Oski said in his preface to the first edition: 'I wish I had written this book'!

Professor Sir Alan Craft
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The contributors to this edition have extensively drawn on the material prepared for previous editions.

Preface

This textbook has been written for undergraduates. Our aim has been to provide the core information required by medical students for the 6–10 weeks assigned to paediatrics in the curriculum of most undergraduate medical schools. We are delighted that it has become so widely used not only in the UK but also in northern Europe, India, Pakistan, Australia, South Africa and other countries. We are also pleased that nurses, therapists and other health professionals who care for children have found the book helpful. It will also be of assistance to doctors preparing for postgraduate examinations such as the Diploma of Child Health (DCH) and Membership of the Royal College of Paediatrics and Child Health (MRCPCH).

The huge amount of positive feedback we have received on the first three editions from medical students, postgraduate doctors and their teachers in the UK and abroad has spurred us on to produce this new edition. We have updated the text and some sections, such as child protection and global health, have been rewritten to incorporate significant advances.

In order to make learning from this book easier, we have followed a lecture-note style using short sentences and lists of important features. Illustrations have been used to help in the recognition of important signs or clinical features and to make the book more attractive and interesting to use. Key learning points are identified and case histories chosen to demonstrate points within their clinical context. Summary boxes of important facts have been included to help with revision.

The male gender for children has been used throughout the book for stylistic simplicity.

We would like to thank all our contributors and Pauline Graham at Elsevier for their assistance in producing this new edition. Thanks also to Ann Goldman, Rachel and David and Sam Lissauer for their ideas and assistance, and for their understanding of the time taken away from the family in the preparation of this new edition.

We welcome any comments about the book.

Tom Lissauer and Graham Clayden

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1

The child in society

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Most medical encounters with children involve an individual child presenting to a doctor with a symptom, such as diarrhoea. After taking a history, examining the child and performing any necessary investigations, the doctor arrives at a diagnosis or differential diagnosis and makes a management plan. This disease-oriented approach, which is the focus of most of this book, plays an important part in ensuring the immediate and long-term well-being of the child. Of course, the doctor also needs to understand the nature of the child's illness within the wider context of their world, which is the primary focus of this chapter. The context of any symptom will affect the:

- likely cause: if the diarrhoea is likely to be from a viral illness or from a contaminated water supply if in a developing country
- severity of the child's illness: the organism likely to be responsible and the child's nutritional status
- management options: who will take care of the child when ill; is pre-prepared oral rehydration therapy available; is hospital treatment possible and what facilities can it offer?

In order to be a truly effective clinician, the doctor must be able to place the child's clinical problems within the context of the family and of the society in which they live.

Important goals for a society are that its children and young people are healthy, safe, enjoy, achieve and make a positive contribution and achieve economic well-being (*Every Child Matters*, 2003 at: http://www.dcsf.gov.uk/everychildmatters). These are included in the UN Rights of the Child (see below). The way in which the environment impacts on a child achieving good health is exemplified by the contrast between the major child health problems in developed and developing countries. In developed countries these are a range of complex, often previously fatal, chronic disorders and behavioural, emotional or developmental problems. By contrast, in developing countries the predominant problems are infection and malnutrition (Box 1.1).

The child's world

Children's health is profoundly influenced by their social, cultural and physical environment. This can be considered in terms of the child, the family and immediate social environment, the local social fabric and the national and international environment (Fig. 1.1). Our ability to intervene as clinicians needs to be seen within this context of complex interrelating influences on health.

The child

The child's world will be affected by gender, genes, physical health, temperament and development. The impact of the social environment varies markedly with age:

- Infant or toddler: life is mainly determined by the home environment
- Young child: by school and friends
- Teenager: also aware of and influenced by events nationally but also internationally, e.g. in music, sport, fashion or politics.

Immediate social environment

Family structure

Although the 'two biological parent family' remains the norm, there are many variations in family structure. In the UK, the family structure has changed markedly over the last 30 years (Fig. 1.2).

Single-parent households – One in four children now live in a single-parent household. Disadvantages of single parenthood include a higher level of unemployment, poor housing and financial hardship (Table 1.1). These social adversities may affect parenting resources, e.g. vigilance about safety, adequacy of nutrition,

Box 1.1 Contrast between main child health problems and associated factors in developed and developing countries

Developed countries – main child health problems	Problems in both developed and developing countries	Developing countries – main child health problems
• Severe, often previously fatal chronic disorders – malignant disease, cystic fibrosis • Provision of paediatric and neonatal intensive care, organ transplantation and other specialist services • Behavioural and emotional disorders – attention deficit hyperactivity disorder, anorexia nervosa • Neurodevelopmental disorders – language delay, reading difficulties, clumsiness, cerebral palsy • Excessive consumption – obesity • Drug and alcohol abuse, smoking, teenage pregnancies.	• Relative socioeconomic disadvantage among the 'have-nots' – lack of money, unemployment, inadequate housing and education • Lack of family cohesion • Healthcare – not available or poor quality or inequality of access	• High mortality rate for children, especially infants • Infection – respiratory tract, diarrhoea, malaria, tuberculosis, HIV • Malnutrition – marasmus, kwashiorkor, severe iron deficiency anaemia • Poor sanitation, water supply, food hygiene • Road traffic and other accidents • Developmental and learning problems of organic pathology – Down syndrome, congenital anomalies • High birth rate – large number of children requiring health care relative to population

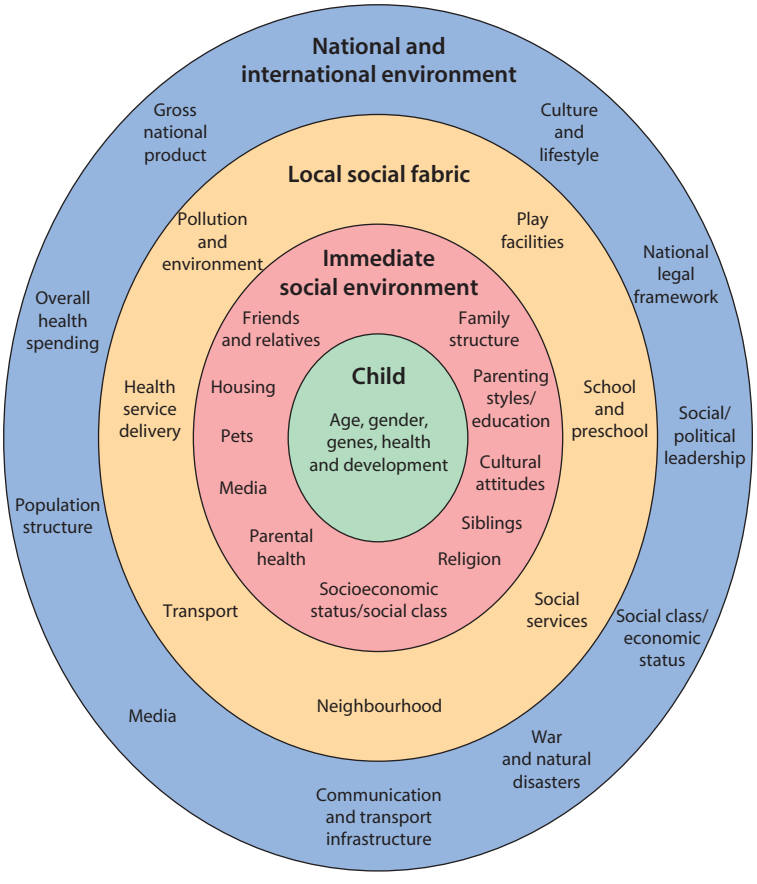


Figure 1.1 A child's world consists of overlapping, interconnected and expanding socioenvironmental layers, which influence children's health and development. (After Bronfenbrenner U. 1979. Contexts of child rearing – problems and prospects. *American Psychologist* 34:844–850.)

Table 1.1 Comparison between parents who are single or couples

	Lone-parent family	Couple family
Median weekly family income (£)	280	573
In lowest income quintile (%)	48	7
Living in social housing (%)	44	12
Parent with no educational qualification (%)	15	3
Child with school behaviour problems (%)	14	8

General Household Survey, Office for National Statistics, England 2008.

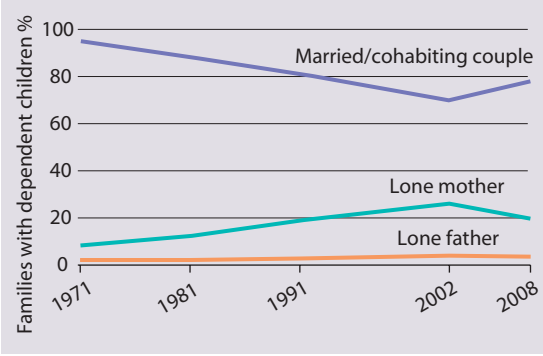


Figure 1.2 Changing structure of the family 1971–2008. (ONS, General Lifestyle Survey 2008).

take-up of preventive services, such as immunisation and regular screening, and ability to cope with an acutely sick child at home.

Reconstituted families – The increase in the number of parents who change partners and the accompanying rise in reconstituted families (1 in 10 children live in a stepfamily) mean that children are having to cope with a range of new and complex parental and sibling relationships. This may result in emotional, behavioural and social difficulties.

Looked after children – Approximately 3% of children under 16 years old in the UK live away from their family home. At any one time in England, over 60 000 children are cared for by Local Authorities. These 'looked after children' are known to have worse outcomes in terms of physical health, mental health, education and employment.

Asylum seekers – These are often placed in temporary housing and moved repeatedly into areas unfamiliar to them. In addition to the uncertainty as to whether or not they will be allowed to stay in the country, they face additional problems as a result of communication difficulties, poverty, fragmentation of families and racism. Many have lost family members and are uncertain about the safety of friends and family.

Parental employment – With many parents in employment, many young children are with childminders or at preschool nurseries. Parents are receiving conflicting opinions on the long-term consequences of caring for their young children at home in contrast to nursery care. Also, increasing attention is being paid to the quality of day-care facilities in terms of supervision of the children and improving the opportunities they provide for social interaction and learning.

Parenting styles

Children rely on their parents to provide love and nurture, stimulation and security, as well as catering for their physical needs of food, clothing and shelter. Parenting that is warm and receptive to the child, while imposing reasonable and consistent boundaries, will promote the development of an autonomous and self-reliant adult. This constitutes 'good enough' parenting as described by the paediatrician and psychotherapist, Donald Winnicott, and can reassure parents that perfection is not necessary. Some parents are excessively authoritarian or extremely permissive. Children's emotional development may be damaged by parents who neglect or abuse their children.

The child's temperament is also important, especially when there is a mismatch with parenting style, for example, a child with a very energetic temperament may be misperceived in a quiet family as having attention deficit hyperactivity disorder (ADHD).

Siblings and extended family

Siblings clearly have a marked influence on the family dynamics. How siblings affect each other appears to be determined by the emotional quality of their relationships with each other and also with other members of the family, including their parents. The arrival of a new baby may engender a feeling of insecurity in older brothers and sisters and result in attention-seeking behaviour. In contrast, children can benefit greatly from having siblings; and from having a close child companion, and can learn from and support each other. The role of grandparents and other family members varies widely and is influenced by the family's culture. In some, they are the main caregivers; in others, they provide valued practical and emotional support. However, in many families they now play only a peripheral role, exacerbated by geographical separation.

Cultural attitudes to child-rearing

The way in which children are brought up evolves within a community over generations, and is influenced by culture and religion, affecting both

day-to-day issues to fundamental lifestyle choices. For example, in some societies children are given considerable self-autonomy, from deciding what food they want to eat, to their education and even to participating in major decisions about their medical care. By contrast, in other societies, children are largely excluded from decision-making. Other examples of marked differences between societies are the use of physical punishment to discipline children and the expected roles of males and females both as children and as adults.

Peers

Peers exert a major influence on children. Peer relationships and activities provide a 'sense of group belonging' and have potentially long-term benefits for the child. Conversely, they may exert negative pressure through inappropriate role modelling. Relationships can also go wrong, e.g. persistent bullying, which may result in or contribute to psychosomatic symptoms, misery and even, in extreme cases, suicide.

Socioeconomic status

Socioeconomic status is a key determinant of health and well-being of children. It is estimated that 2.8 million children in the UK are living in poverty (below 60% of the national median income after adjustment for housing costs). The proportion of children in poverty in different countries is shown in Figure 1.3. Health issues in the UK in which prevalence rates are increased by poverty include:

- low birthweight infants
- injuries
- hospital admissions
- asthma
- behavioural problems
- special educational needs
- child abuse.

Low socioeconomic status is often associated with multiple disadvantages, e.g. food of inadequate quantity or poor nutritional value, substandard housing or homelessness, lack of 'good enough' parenting, parental education and health, and poor access to healthcare and educational facilities.

There are marked differences in living experiences between ethnic groups: 50% of Afro-Caribbean children live in single-parent households compared with 15% of white children and less than 10% of those from the Indian subcontinent. In 1992, in England and Wales, 12% of births were to mothers born outside the UK; in 2008 it was nearly 24%.

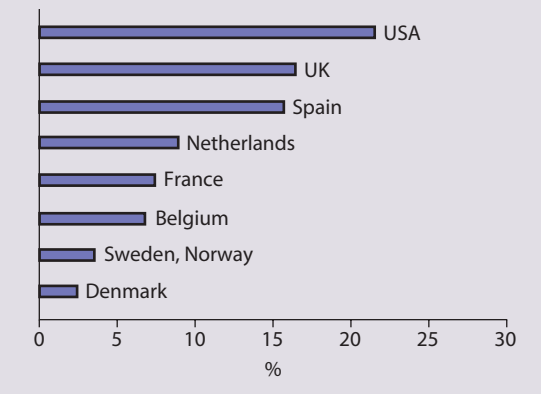


Figure 1.3 Percentage of children living in poverty. UNICEF definition is households with income below 50% of national median (UNICEF 2009). (Innocenti Research Centre. 2000. *British Medical Journal* 320:1621, reproduced with permission from the BMJ publishing group.)

Local social fabric

Neighbourhood

Cohesive communities and amicable neighbourhoods are positive influences on children. Racial tension and other social adversities, such as gang violence and drugs, will adversely affect the emotional and social development of children, as well as their physical health. Parental concern about safety may create tensions in balancing their children's freedom with overprotection and restriction of their lifestyles. The physical environment itself, through pollution, safe areas for play and quality of housing and public facilities, will affect children's health.

Health service delivery

The variation in the quality of healthcare is an important component in preventing morbidity and mortality in children. Health services for children are increasingly provided within primary care. Some aspects of specialist paediatric care are also increasingly provided within the child's home, local community or local hospital through shared care arrangements and specialist community nursing and medical teams working within clinical networks. However, access to and the range of these services varies widely.

Schools

Schools provide a powerful influence on children's emotional and intellectual development and their subsequent lives. Differences in the quality of schools in different areas can accentuate inequalities already present in society. Schools provide enormous opportunities for influencing healthy behaviour through personal and social education and through the influence of peers and positive role models. They also provide opportunities for monitoring and promoting the health and well-being of vulnerable children.

To educate a girl is to educate a whole family. And what is true of families is also true of communities and, ultimately, whole countries. Study after study has taught us that there is no tool for development more effective than the education of girls. No other policy is as likely to raise economic productivity, lower infant and maternal mortality, improve nutrition and promote health (Kofi Annan, Secretary General UN 2004).

Travel

The increasing ease of travel can broaden children's horizons and opportunities. Especially in rural areas, the ease and availability of transport allow greater access to medical care and other services. However, the increasing use of motor vehicles contributes to the large number of injuries sustained by children from road traffic accidents, mainly as pedestrians. It also decreases physical activity, as shown by the high proportion of children taken to school by car. Whereas 80% of children in the UK went to school by foot or bicycle in 1971, this has dropped to less than 10%.

National and international environment

Economic wealth

In general, there is an inverse relationship between a country's gross national product and income distribution and the quality of its children's health.

The lower the gross national income:

- the greater the proportion of the population who are children
- the higher the childhood mortality
- the higher the proportion of newborn infants with low birthweight
- the lower the immunisation rate.

However, even in countries with a high gross national product, many children live in financially deprived circumstances.

The dramatic reduction in childhood mortality in England over the last century is shown in Figure 1.4. It is primarily from improvements in living conditions such as improved sanitation and housing, and access to food and water. These have dramatically reduced fatalities from infectious disease. More recent contributions to this reduction have been increased availability and uptake of immunisation and major medical improvements in perinatal and infant care.

In all countries difficult choices need to be made about the allocation of scarce resources. Should a developing country provide very expensive drugs and care for the small number of children with malignant disease or allocate its resources to preventive programmes for many children? In developed countries, difficult decisions also have to be faced in deciding the affordability of very expensive procedures, such as heart or liver transplantation, or neonatal intensive care for extremely premature infants and certain drugs, such as the genetically engineered enzyme replacement therapy for Gaucher disease or cytokine modulators ('biologics') and other immunotherapies. The public are becoming more engaged in these debates.

Media and technology

The media has a powerful influence on children. It can be positive and educational. However, the impact of television and computers and mobile technology can be negative owing to reduced opportunities for social interaction and active learning, lack of physical exercise and exposure to violence, sex and cultural stereotypes. The extent to which the aggressive tendencies of children may be exacerbated or encouraged by media exposure to violence is unclear.

The internet is enabling parents and children to become better informed about and gain support for their children's medical problems. This is especially beneficial for the many rare conditions encountered in paediatrics. A disadvantage is that it may result in the dissemination of information which is incorrect or biased, and may result in requests for inappropriate or untested investigations or treatment.

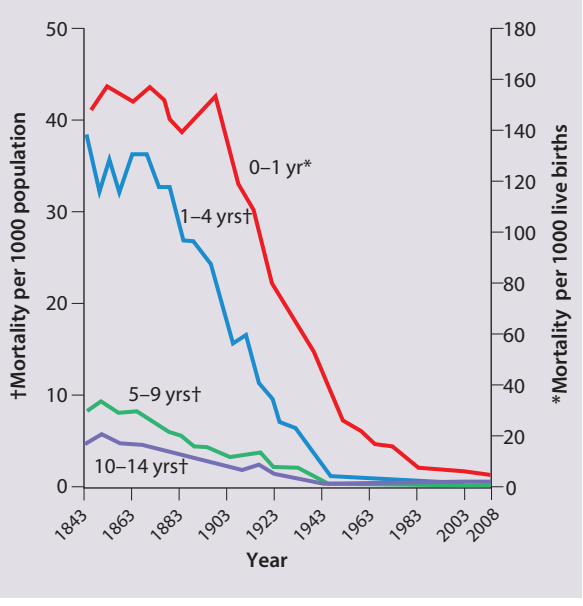


Figure 1.4 Marked reduction in mortality of children aged 0–14 years between 1843 and 2008. In 1900, 15% of babies born in England died by 1 year of age and 23% by 14 years of age; in 2008 the figures were 0.46% and 0.64%, respectively.

Box 1.2 Children and war: worldwide, devastating effect of war on children in the last decade

- Mortality: >2 million children died
- Anti-personnel mines: 8000–10 000 killed or maimed each year
- Morbidity: >6 million children disabled, mainly paraplegia and sensory deficits
- Loss of home and refugee status: 20 million children homeless and living as refugees
- Orphans: 1 million children orphaned
- Psychological trauma: 10 million children estimated to have post-traumatic stress syndrome; rape and sexual humiliation of females widely used as a strategy of conflict
- Children as soldiers: estimated 300 000 child soldiers in more than 30 conflicts worldwide
- Disruption of healthcare system: immunisation and child health surveillance programmes interrupted or disbanded.

Source: State of World's Children, UNICEF 2005.

War and natural disasters

Children are especially vulnerable when there is war, civil unrest or natural disasters (Box 1.2). Not only are they at greater risk from infectious diseases and malnutrition but also they may lose their caregivers and other members of their families and are likely to have been exposed to highly traumatic events.

Their lives will have been uprooted, socially and culturally, especially if they are forced to flee from their homes and become refugees.

Public health issues for young people

Important public health issues for the 11 million children and young people in the UK in which doctors play a role include:

Obesity – it is estimated that 6.5% of 9-year-olds and 15% of 15-year-olds are clinically obese (BMI >90th centile). Doctors can help promote healthy eating through supporting breast-feeding in infancy, advising parents and young people on healthy lifestyles, monitoring growth parameters and the consequences of obesity, and through advocacy and support for local and national healthy lifestyle programmes.

Emotional and behavioural difficulties – 11% of boys and 8% of girls in this country suffer from a defined emotional or behavioural difficulty. In addition, these problems are often unrecognised and have significant ongoing impacts on children's overall well-being. Doctors can contribute to tackling these problems by being alert to and responding to the signs of mental health problems in childhood, and through promoting equitable distribution of resources to child and adolescent mental health services.

Disability – up to 5.4% have some form of disability and 7% have a long-standing illness that limits their activity. Doctors need to work closely with children and young people, families, local communities and other services to ensure that any individual child's needs are appropriately catered for. This may include outlining a child's health needs for a statement of special educational need, formulating an individual healthcare plan and advocating for the resources to implement this. Doctors can also provide education and social services with data on the numbers and levels of need within their own population.

Smoking, alcohol and drugs – a 2007 survey found that 6% of 11–15-year-olds smoke regularly; 10% had taken drugs in the past month and 20% had drunk alcohol in the past week. Doctors have been instrumental in campaigning for legislation to protect young people from targeted advertising and to raise awareness of the dangers of smoking, alcohol and drugs. There is some evidence that prevalence of all three behaviours are decreasing.

Doctors can also help children through advocacy about children's issues, by providing information to inform public debate. Examples of this are child protection, exploitation of children for labour or trafficking, the needs of refugee children and tobacco and alcohol advertising.

Children's rights

Children's rights are laid down in the United Nations Convention on the Rights of the Child, which has been ratified by all members of the United Nations, including the UK, but excepting the USA and Somalia (Box 1.3, Fig. 1.5). Unfortunately, the rights of many children are not met. Implications of the convention include the involvement of children in clinical decision-making and in issues of consent.

Box 1.3 Summary of the United Nations Convention on the Rights of the Child (1989)

1. Survival rights

The child's right to life and to the most basic needs – food, shelter and access to healthcare.

2. Developmental rights

To achieve their full potential – education, play, freedom of thought, conscience and religion. Those with disabilities to receive special services.

3. Protection rights

Against all forms of abuse, neglect, exploitation and discrimination.

4. Participation rights

To take an active role in their communities and nations.

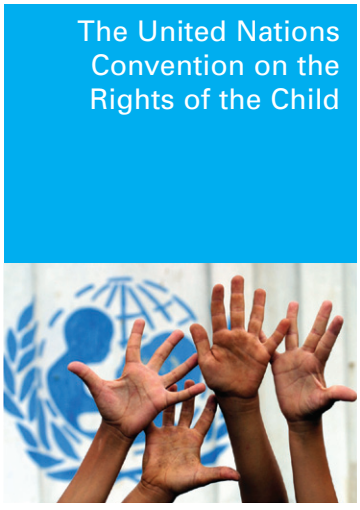


Figure 1.5 United Nations Convention On the Rights of the Child (1989). Reproduced with the permission of UNICEF.

Summary

Regarding child and young people's health

- For the individual child: depends on the child, the immediate social environment (family, parenting, peers, socioeconomic status), the local social fabric and the national and international environment
- The main problems in developed countries are chronic medical disorders, behavioural and emotional disorders, neurodevelopmental disorders, accidents, socioeconomic disadvantage, excessive consumption of food, drug and alcohol abuse, smoking and teenage pregnancies.

Global child health

Child mortality

Worldwide, each year, 8.8 million children under 5 years old die (Fig. 1.6). Infectious diseases, with undernutrition a major contributing factor, account for most deaths in children. Pneumonia and diarrhoea are the leading infectious causes of death. More than half of child deaths occurring after the neonatal period are due to just five preventable and treatable conditions: pneumonia, diarrhoea, malaria, measles and HIV.

Neonatal mortality (first 4 weeks of life) accounts for 41% of all under-5 deaths. Maternal health and care, especially at delivery, plays a crucial role. Preterm birth, infections and birth asphyxia are the leading causes of neonatal death; being underweight is an important contributing factor.

Most child deaths in developing countries are preventable.

Where deaths occur

Not surprisingly, the mortality for infants and children living in poor countries is much higher than for those in rich countries. The lethal combination of poverty, HIV/AIDS and armed conflict underlies the very high under-5 mortality in some countries. In addition, many poor countries have warm, humid climates where tropical diseases such as malaria occur. However, even in these countries, much of the excess disease burden is due to illnesses such as respiratory infections and diarrhoea that occur at a much greater frequency and severity than in developed countries.

Although only about 48% of the world's estimated 629 million under-5s live in sub-Saharan Africa and

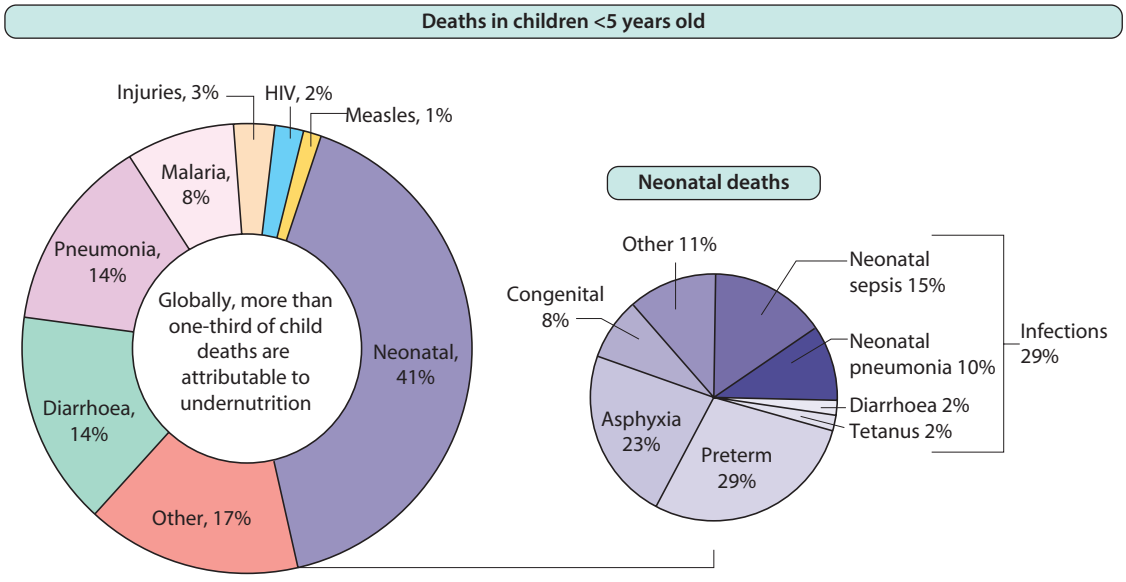


Figure 1.6 Causes of the 8.8 million deaths among children under 5 years old, 2008. (WHO and UNICEF 2010. http://www.childinfo.org. Reproduced with permission. Accessed January 2011).

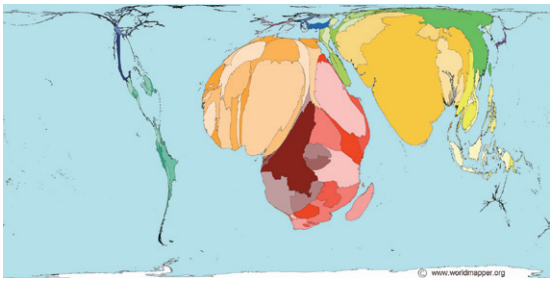


Figure 1.7 Territory size shows the proportion of all deaths of children aged 1–5 years. (http://www.worldmapper.org, 2002 data. Accessed January 2011).

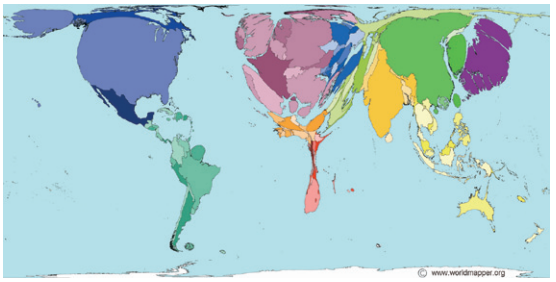


Figure 1.8 Territory size shows the proportion of doctors that work in that territory. (http://www.worldmapper.org, 2002 data. Accessed January 2011).

South Asia, 93% of child deaths occur in these two regions (Fig. 1.7). The difference in resources available is reflected in the difference in the number of healthcare professionals in different parts of the world. This is shown in diagrammatic form in Figure 1.8. Whereas the USA spends over $6000/person/year on health, the annual expenditure is only $25/person in the 49 lowest income countries. Put another way, the annual health budget available to the 1.2 billion people living in these countries is equivalent to that for the 6.5 million living in Arizona!

Over half of all child deaths occur in just five countries, which have high mortality rates and large populations: India, Nigeria, Democratic Republic of Congo (DRC), Pakistan and China. Nine countries have an under-5 years mortality rate of ≥180/1000 live births (Sierra Leone, Afghanistan, Chad, Equatorial Guinea, Guinea-Bissau, Mali, Burkina Faso, Nigeria, Burundi). This means that a child's chance of dying before age 5 years is 30-fold greater in these countries than in developed countries, where the under-5 years mortality rate is 6/1000 live births. The impact of poverty and limited health facilities on children is demonstrated by Case History 1.1.

Reducing child mortality

The international community has taken many steps to improve child health in poor countries. Some are listed in Box 1.4.

The launch of the Integrated Management of Childhood Illness (IMCI) in 1992 was a major advance (Fig. 1.10). It recognised that an integrated management approach to childhood illness is required, including nutrition and preventative care in families and communities. In health facilities, sick children are triaged according to the presence of specific danger signs and management is planned according to algorithms which can be followed by non-medical staff.

The major advance in global child health has been in the adoption of the Millennium Development Goals (MDGs), which has served as a major focus for the international community's commitment to reduce child mortality (Fig. 1.11). There are specific targets for each goal. For MDG 4 it is to reduce child mortality in children <5 years by two-thirds, between the years 2000

Case History

1.1 Burns to the face

This girl in West Africa sustained facial burns (Fig. 1.9). She fell into the open kitchen fire when she had a seizure. Although her epilepsy had initially responded well to phenobarbital, regular supplies were unavailable. Difficulty in finding affordable transport from the village to the health clinic delayed presentation by 4 days, by which time there was secondary infection and increased risk of cataract from conjunctival injury.

This simple example highlights the influence of environment on children's health. Her illness was readily treatable, her injuries were preventable with a fireguard, delayed treatment resulted in complications and only basic medical care was available at the clinic.



Figure 1.9 Facial burn in West Africa.

Box 1.4 Key international developments in maternal and child health

- 1946: UNICEF established from United Nations Emergency Fund
- 1948: WHO was formed
- 1974: Extended Programme of Immunisation (EPI) included diphtheria, pertussis, tetanus, polio, TB and measles
- 1978: Alma Ata. Primary Healthcare – Health for All
- 1980s: GOBI-FFF. Key primary care strategies: growth monitoring, oral rehydration therapy, breast-feeding, immunisation, family spacing, female education and food supplementation
- 1989: Convention on the Rights of the Child
- 1992: World Summit for Children – Child-to-Child strategy, Baby Friendly initiative, Integrated Management of Childhood Illness (IMCI)
- 2000: United Nations endorses Millennium Development Goals.

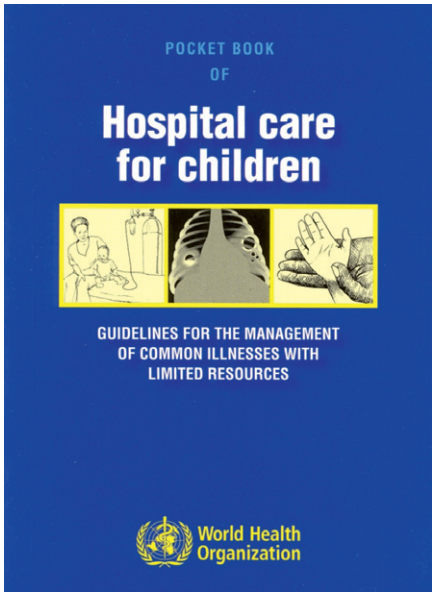


Figure 1.10 Integrated Management of Childhood Illness (IMCI) guidelines for countries with limited resources, WHO. Reproduced with permission.



Figure 1.11 Millennium Development Goals. (Available at: http://www.undp.rog/mdg. Accessed January 2011). UNDP Brazil. Reproduced with permission.

and 2015. Contributions are also made by MDGs to improve maternal health and reduce mortality from malaria by insecticide-treated bednets and appropriate anti-malarial drugs and from HIV and TB.

Some of the specific health measures targeted in maternal and child health are shown in Figure 1.12, together with the overall success in implementing them.

The health measures include:

- Contraception available
- One or more antenatal visits
- Skilled attendant at delivery
- Postnatal visit within 2 days
- Exclusive breast-feeding
- Case management of pneumonia
- Measles immunisation.

The aim is for 100% coverage.

Progress in reducing under-5-year-old mortality is being made (Fig. 1.13), although in many of the poorest countries it is slow and patchy, and considerable effort will be required to meet the MDG 4 target.

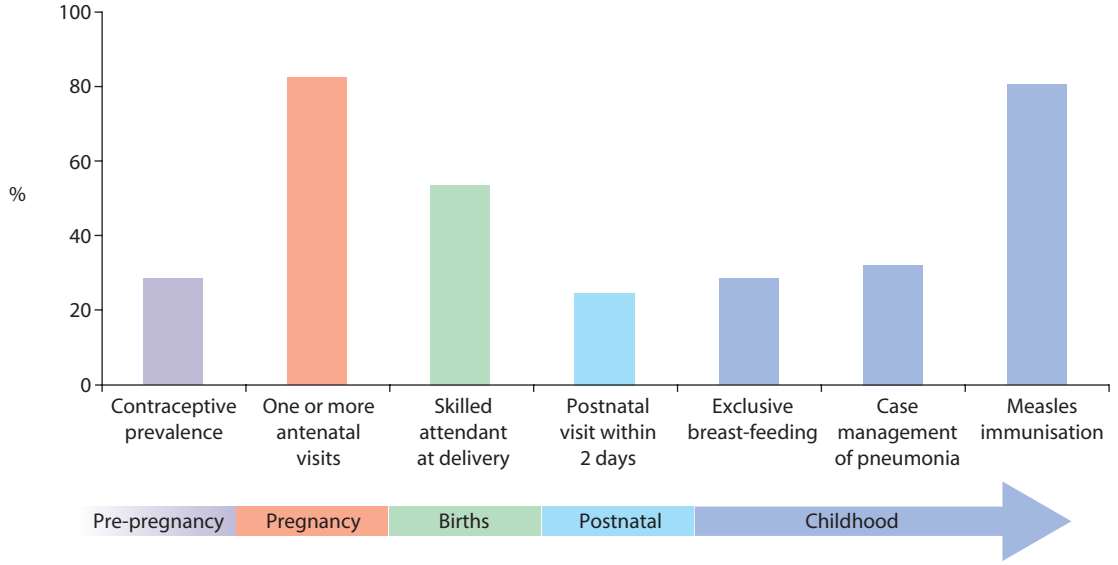


Figure 1.12 Coverage estimates for interventions across the continuum of care (2000–2006) (Countdown Coverage Writing Group; Countdown to 2015 Core Group, Bryce J, Daelmans B, Dwivedi A et al. 2008. Countdown to 2015 for maternal, newborn, and child survival: the 2008 report on tracking coverage of interventions. *Lancet* 371:1247–1258).

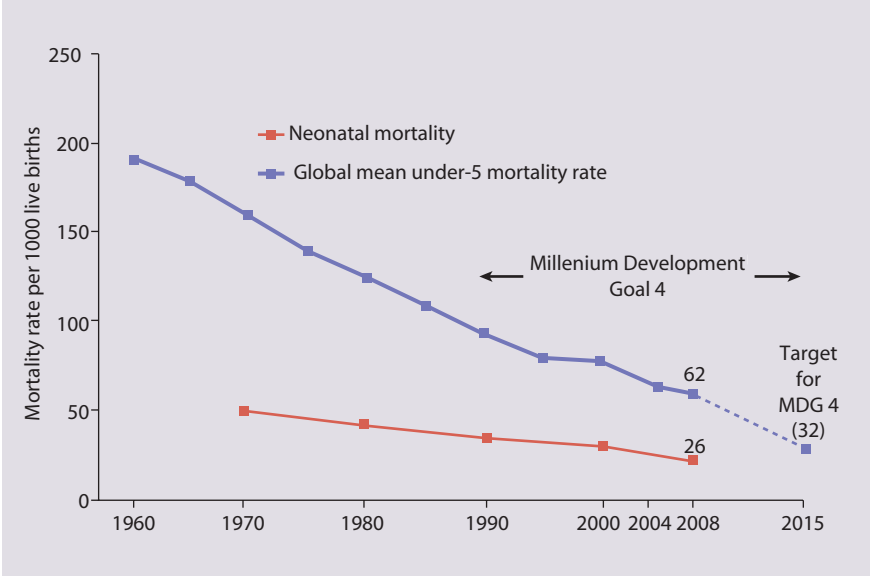


Figure 1.13 Global progress towards Millennium Development Goal 4 (MDG 4) for child survival (Adapted from Lawn J, Newborn survival in low resource settings – are we delivering? *BJOG* 116 (Suppl 1):49–59, 2009. updated 2010 for data to 2008).

Conclusion

Children are vulnerable members of society. They rely on their parents and society to care for them and provide an environment where they can grow both physically and emotionally to reach their full potential. Their health is dependent on a nurturing environment and good health services. Where these are deficient, whether in developed or developing countries, children suffer, but in developing countries they not infrequently lead to death. A global child health perspective empowers the student or child health professional to contribute to the global health workforce, join in international education and research and, most importantly, serve as an effective advocate for the world's children.

Further reading

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2

History and examination

Taking a history	13	Communicating with children	27
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Examination	17		

The cornerstone of clinical practice continues to be history-taking and clinical examination. Good doctors will continue to be admired for their ability to distil the important information from the history, for their clinical skills, for their attitude towards patients and for their knowledge of diseases, disorders and behaviour problems.

Parents are acutely interested in and anxious about their children. They will quickly recognise doctors who demonstrate interest, empathy and concern. They will seek out doctors who possess the appropriate skills and attitudes towards their children.

In approaching clinical history and examination of children, it is helpful to visualise some common clinical scenarios in which children are seen by doctors:

- An acute illness, e.g. respiratory tract infection, meningitis, appendicitis
- A chronic problem, e.g. failure to thrive, chronic cough
- A newborn infant with a congenital malformation or abnormality, e.g. developmental dysplasia of the hip, Down syndrome
- Suspected delay in development, e.g. slow to walk, talk or acquire skills
- Behaviour problems, e.g. temper tantrums, hyperactivity, eating disorders.

The aims and objectives are to:

- establish the relevant facts of the history; this is always the most fruitful source of diagnostic information
- elicit all relevant clinical findings
- collate the findings from the history and examination
- formulate a working diagnosis or differential diagnosis on the basis of logical deduction
- assemble a problem list and management plan.

The above can be summarised by the acronym HELP:

H = history
E = examination
L = logical deduction
P = plan of management.

Key points in paediatric history and examination are:

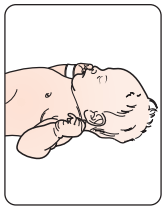
- the child's age – a key feature in the history and examination (Fig. 2.1) as it determines:
  - the nature and presentation of illnesses, developmental or behaviour problems
  - the way in which the history-taking (Fig. 2.2) and examination are conducted
  - the way in which any subsequent management is organised
- the parents, who are astute observers of their children. Never ignore or dismiss what they say.

Taking a history

Introduction

- Make sure you have read any referral letter and scanned the notes *before* the start of the interview.
- Observe the child at play in the waiting area and observe their appearance, behaviour and gait as they come into the clinic room. The continued observation of the child during the whole interview may provide important clues to the diagnosis and management.
- When you welcome the child, parents and siblings, check that you know the child's first name and gender. Ask how the child prefers to be addressed.

Paediatrics is a specialty governed by age



Infant

Neonate (<4 weeks)
Infant (<1 year)



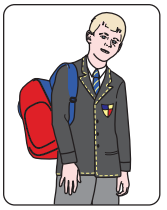
Toddler

Approx 1–2 years



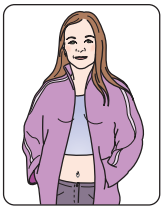
Preschool

Young child (2–5 years)



School-age

Older child



Teenager

Adolescent

Figure 2.1 The illnesses and problems children encounter are highly age-dependent. The child's age will determine the questions you ask on history-taking; how you conduct the examination; the diagnosis or differential diagnosis and your management plan.

Paediatrics stretches from newborn infants to adolescents. Whenever you consider a paediatric problem, whether medical, developmental or behavioural, first ask, 'What is the child's age?'

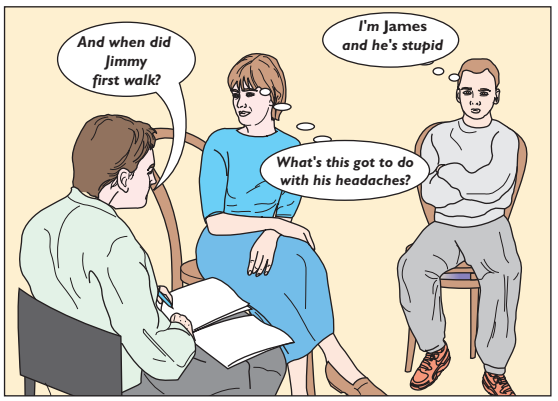


Figure 2.2 The history must be adapted to the child's age. The age when a child first walks is highly relevant when taking the history of a toddler but irrelevant for a teenager with headaches.

- Introduce yourself.
- Determine the relationship of the adults to the child.
- Establish eye contact and rapport with the family. Infants and some toddlers are most secure in parents' arms or laps. Young children may need some time to get to know you.
- Ensure that the interview environment is as welcoming and unthreatening as possible. Avoid having desks or beds between you and the family, but keep a comfortable distance.
- Have toys available. Observe how the child separates, plays and interacts with any siblings present.
- Do not forget to address questions to the child, when appropriate.
- There will be occasions when the parents will not want the child present or when the child should be seen alone. This is usually to avoid embarrassing older children or teenagers or to impart sensitive information. This must be handled tactfully, often by negotiating to talk separately to each in turn.

Presenting symptoms

Full details are required of the presenting symptoms. Let the parents and child recount the presenting complaints in their own words and at their own pace. Note the parent's words about the presenting complaint: onset, duration, previous episodes, what relieves/aggravates them, time course of the problem, if getting worse and any associated symptoms. Has the child's or the family's lifestyle been affected? What has the family done about it?

Make sure you know:

- What prompted referral to a doctor
- What the parents think or fear is the matter.

The scope and detail of further history-taking are determined by the nature and severity of the presenting complaint and the child's age. While the comprehensive assessment listed here is sometimes required, usually a selective approach is more appropriate (Fig. 2.3). This is not an excuse for a short, slipshod history, but instead allows one to focus on the areas where a thorough, detailed history is required.

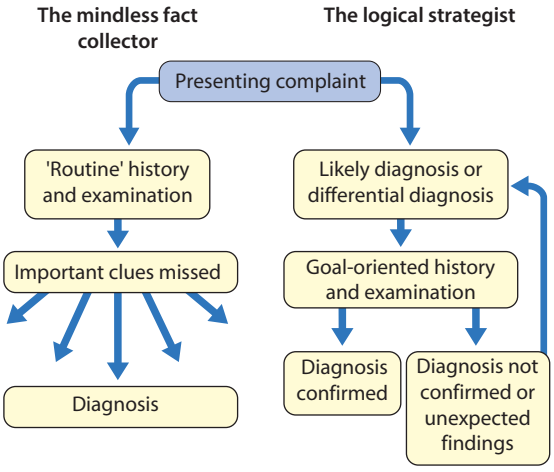


Figure 2.3 The history and examination should be goal-oriented, based on the presenting complaint. Comprehensive history-taking is best reserved for training or for complex, multi-system disorders. (Adapted from Hutson JM, Beasley SW. 1988. *The Surgical Examination of Children*. Heinemann Medical Books, London.)

General enquiry

Check:

- General health – how active and lively?
- Normal growth
- Pubertal development (if appropriate)
- Feeding/drinking/appetite
- Any recent change in behaviour or personality.

Systems review

Selected, as appropriate:

- General rashes, fever (if measured)
- Respiratory – cough, wheeze, breathing problems
- ENT – throat infections, snoring, noisy breathing (stridor)
- Cardiovascular – heart murmur, cyanosis, exercise tolerance
- Gastrointestinal – vomiting, diarrhoea/constipation, abdominal pain
- Genitourinary – dysuria, frequency, wetting, toilet-trained
- Neurological – seizures, headaches, abnormal movements
- Musculoskeletal – disturbance of gait, limb pain or swelling, other functional abnormalities.

Make sure that you and the parent or child mean the same thing when describing a problem.

Past medical history

Check:

- Maternal obstetric problems, delivery, prolonged rupture of membranes, group B streptococcus status, maternal pyrexia (pertinent if neonate)
- Birthweight and gestation
- Perinatal problems, whether admitted to special care baby unit, jaundice, etc.
- Immunisations (ideally from the personal child health record)
- Past illnesses, hospital admissions and operations, accidents and injuries.

Medication

Check:

- Past and present medications
- Known allergies.

Family history

Families share houses, genes and diseases!

- Have any members of the family or friends had similar problems or any serious disorder?

Any neonatal/childhood deaths?

- Draw a family tree. If there is a positive family history, extend family pedigree over several generations.
- Is there consanguinity?

Social history

Check:

- Relevant information about the family and its community – parental occupation, economic status, housing, relationships, parental smoking, marital stresses.
- Is the child happy at home? What are the child's preferred play or leisure activities?
- Is the child happy at nursery/school?
- Is there a social worker involved?

This 'social snapshot' is crucial, since many childhood illnesses or conditions are permeated by adult problems, for example:

- alcohol and drug abuse
- long-term unemployment/poverty
- poor, damp, cramped housing
- parental psychiatric disorders
- unstable partnership.

Development

Check:

- Parental worries about vision, hearing, development
- Key developmental milestones (Fig. 2.4)
- Previous child health surveillance developmental checks
- Bladder and bowel control
- Child's temperament, behaviour
- Sleeping problems
- Concerns and progress at nursery/school.

Look through the personal child health record.

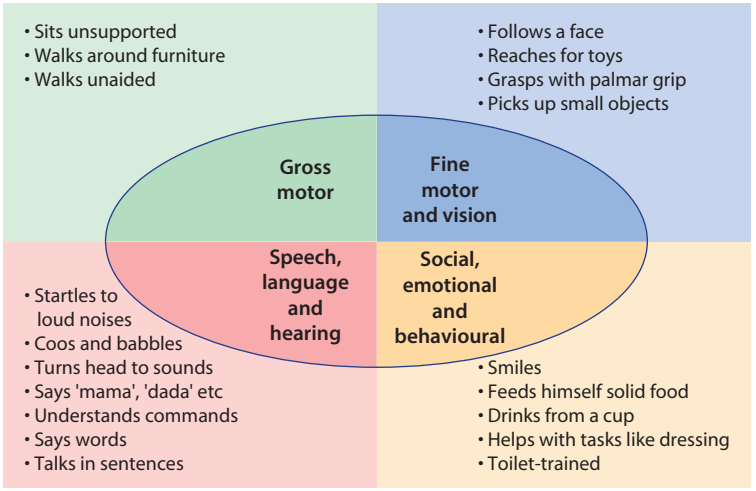


Figure 2.4 Some key developmental milestones in infants and young children. These are considered in detail in Chapter 3.

An approach to examining children

Obtaining the child's cooperation

- Make friends with the child.
- Be confident but gentle.
- Avoid dominating the child.
- Short mock examinations, e.g. auscultating a teddy or the mother's hand, may allay a young child's fears.
- When first examining a young child, start at a non-threatening area, such as a hand or knee.
- Explain what you are about to do and what you want the child to do, in language he can understand. As the examination is essential, not optional, it is best not to ask his permission, as it may well be refused!
- A smiling, talking doctor appears less threatening, but this should not be overdone as it can interfere with one's relationship with the parents.
- Leave unpleasant procedures until last.

Adapting to the child's age

Adapt the examination to suit the child's age. While it may be difficult to examine some toddlers and young children fully, it is usually possible with resourcefulness and imagination on the doctor's part.

- Babies in the first few months are best examined on an examination couch with a parent next to them.
- A toddler is best initially examined on his mother's lap or occasionally over a parent's shoulder.
- Parents are reassuring for the child and helpful in facilitating the examination if guided as to what to do (Fig. 2.5).
- Preschool children may initially be examined while they are playing.
- Older children and teenagers are often concerned about privacy. Teenage girls should normally be examined in the presence of their mother, or a nurse or suitable chaperone. Be aware of cultural sensitivities in different ethnic groups.



Figure 2.5 Distracting a toddler with a toy allows auscultation of the heart.

Undressing children

Be sensitive to children's modesty. The area to be examined must be inspected fully but this is best done in stages, re-dressing the child when each stage has been completed. It is easiest and kindest to ask the child or parent to do the undressing.

Warm, clean hands

Hands must be washed before (and after) examining a child. Warm smile, warm hands and a warm stethoscope all help.

Developmental skills

A good overview of developmental skills can be obtained by watching the child play. A few simple toys, such as some bricks, a car, doll, ball, pencil and

paper, pegboard, miniature toys and a picture book, are all that is required, as they can be adapted for any age. If developmental assessment (see Ch. 3) is the focus of the examination, it is advisable to assess this before the physical examination, as cooperation may then be lost.

Examination

Initial observations

Careful observation is usually the key to success in examining children. Look before touching the child. Inspection will provide information on:

- severity of illness
- growth and nutrition
- behaviour and social responsiveness
- level of hygiene and care.

Severity of illness

Is the child sick or well? If sick, how sick? For the acutely ill infant or child, perform the '60-second rapid assessment':

- Airway and Breathing – respiration rate and effort, presence of stridor or wheeze, cyanosis
- Circulation – heart rate, pulse volume, peripheral temperature, capillary refill time
- Disability – level of consciousness.

The care of the seriously ill child is described in Chapter 6.

Measurements

As abnormal growth may be the first manifestation of illness in children, always measure and plot growth on centile charts for:

- weight, noting previous measurements from personal child health record
- length (in infants, if indicated) or height in older children
- head circumference in infants.

Also, as appropriate:

- temperature
- blood pressure
- peak expiratory flow rate.

General appearance

The face, head, neck and hands are examined. The general morphological appearance may suggest a chromosomal or dysmorphic syndrome. In infants, palpate the fontanelle and sutures.

Respiratory system

Cyanosis

Central cyanosis is best observed on the tongue.

Clubbing of the fingers and/or toes

Clubbing (Fig. 2.6a) is usually associated with chronic suppurative lung disease, e.g. cystic fibrosis, or cyanotic congenital heart disease. It is occasionally seen in inflammatory bowel disease or cirrhosis.

Tachypnoea

Rate of respiration is age-dependent (Table 2.1).

Dyspnoea

Laboured breathing. Increased respiratory rate (may be the only sign of increased work of breathing).

Increased work of breathing is judged by:

- nasal flaring
- expiratory grunting – to increase positive end-expiratory pressure
- use of accessory muscles, especially sternomastoids
- retraction (recession) of the chest wall, from use of suprasternal, intercostal and subcostal muscles
- difficulty speaking (or feeding).

Chest shape

- Hyperexpansion or barrel shape (Fig. 2.6b), e.g. asthma
- *Pectus excavatum* (hollow chest) or *pectus carinatum* (pigeon chest)
- Harrison's sulcus (indrawing of the chest wall from diaphragmatic tug), e.g. from poorly controlled asthma
- Asymmetry of chest movements.

Palpation

- Chest expansion: this is 3–5 cm in school-aged children. Measure maximal chest expansion with a tape measure. Check for symmetry.
- Trachea: checking that it is central is seldom helpful and is disliked by children. To be done selectively.
- Location of apex beat to detect mediastinal shift.

Percussion

- Needs to be done gently, comparing like with like, using middle fingers.
- Seldom informative in infants.
- Localised dullness: collapse, consolidation, fluid.

Auscultation (ears and stethoscope)

- Note quality and symmetry of breath sounds and any added sounds
- Harsh breath sounds from the upper airways are readily transmitted to the upper chest in infants
- Hoarse voice – abnormality of the vocal cords
- Stridor – harsh, low-pitched, mainly inspiratory sound from upper airways obstruction

Respiratory system



Figure 2.6a Clubbing of the fingers. There is increased curvature, loss of nail angle and fluctuation. This child had cystic fibrosis.

Table 2.1 Respiratory rate in children (breaths/min)

Age	Normal	Tachypnoea
Neonate	30–50	>60
Infants	20–30	>50
Young children	20–30	>40
Older children	15–20	>30



Figure 2.6b Hyperexpanded chest from chronic obstructive airways disease. This boy had severe asthma.

Table 2.2 Chest signs of some common chest disorders of children

	Chest movement	Percussion	Auscultation
Bronchiolitis	Laboured breathing Hyperinflated chest Chest recession	Hyper-resonant	Fine crackles in all zones Wheezes may/may not be present
Pneumonia	Reduced on affected side Rapid, shallow breaths	Dull	Bronchial breathing Crackles
Asthma	Reduced but hyperinflated Use of accessory muscles Chest wall retraction	Hyper-resonant	Wheeze

Infants with pneumonia may not have any abnormal signs on auscultation.

Sputum is rarely produced by children, as they swallow it. The main exception is suppurative lung disease from cystic fibrosis.

- Breath sounds – normal are vesicular; bronchial breathing is higher-pitched and the length of inspiration and expiration equal
- Wheeze – high-pitched, expiratory sound from distal airway obstruction (Table 2.2)
- Crackles – discontinuous 'moist' sounds from the opening of bronchioles (Table 2.2).

Cardiovascular system

Cyanosis

Observe the tongue for central cyanosis.

Clubbing of fingers or toes

Check if present.

Pulse

Check:

- Rate (Table 2.3)
- Rhythm – sinus arrhythmia (variation of pulse rate with respiration) is normal
- Volume – small in circulatory insufficiency or aortic stenosis; increased in high-output states (stress, anaemia); collapsing in patent ductus arteriosus, aortic regurgitation.

Inspection

Look for:

- respiratory distress
- precordial bulge – caused by cardiac enlargement

Table 2.3 Normal resting pulse rate in children

Age	Beats/min
<1 year	110–160
2–5 years	95–140
5–12 years	80–120
>12 years	60–100

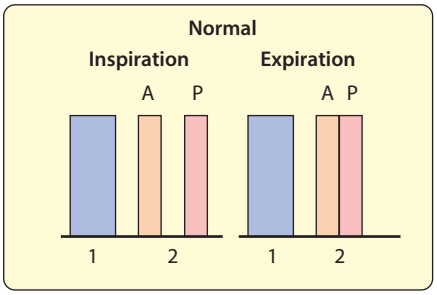


Figure 2.7 The splitting of the second heart sound is easily heard in children.

Features of heart failure in infants:
- **Poor feeding/failure to thrive**
- **Sweating**
- **Tachypnoea**
- **Tachycardia**
- **Gallop rhythm**
- **Cardiomegaly**
- **Hepatomegaly.**

Features suggesting that a murmur is significant:
- **Conducted all over the precordium**
- **Loud**
- **Thrill (equals grade 4–6 murmur)**
- **Any diastolic murmur**
- **Accompanied by other abnormal cardiac signs.**

- ventricular impulse – visible if thin, hyperdynamic circulation or left ventricular hypertrophy
- operative scars – mostly sternotomy or left lateral thoracotomy.

Palpation

Thrill = palpable murmur.

Apex (4th–5th intercostal space, mid-clavicular line):

- not palpable in some normal infants, plump children or dextrocardia
- heave from left ventricular hypertrophy.

Right ventricular heave at lower left sternal edge – right ventricular hypertrophy.

Percussion

Cardiac border percussion is rarely helpful in children.

Auscultation

Listen for heart sounds and murmurs.

Heart sounds

- Splitting of second sound is usually easily heard and is normal (Fig. 2.7).
- Fixed splitting of second heart sound in atrial septal defects.
- Third heart sound in mitral area is normal in young children.

Murmurs

- Timing – systolic/diastolic/continuous
- Duration – mid-systolic (ejection)/pansystolic
- Loudness – systolic murmurs graded:
  - 1–2: soft, difficult to hear
  - 3: easily audible, no thrill
  - 4–6: loud with thrill
- Site of maximal intensity – mitral/pulmonary/aortic/tricuspid areas
- Radiation:
  - to neck in aortic stenosis
  - to back in coarctation of the aorta or pulmonary stenosis.

Draw your findings (see Ch. 17 on cardiac disorders).

Hepatomegaly

Important sign of heart failure in infants. An infant's liver is normally palpable 1–2 cm below the costal margin.

Femoral pulses

In coarctation of the aorta:

- Decreased volume or may be impalpable in infants
- Brachiofemoral delay in older children.

Blood pressure (see later in chapter)

Heart disease is more common in children with other congenital abnormalities or syndromes, e.g. Down and Turner syndromes.

Abdomen

Abdominal examination is performed in three major clinical settings:

- The routine part of the examination
- An 'acute abdomen' – ?cause (see Ch.13, Gastroenterology)
- Recurrent abdominal pain/distension/constipation mass.

Associated signs

Examine:

- The eyes for signs of jaundice and anaemia
- The tongue for coating and central cyanosis
- The fingers for clubbing.

Inspection

The abdomen is protuberant in normal toddlers and young children. The abdominal wall muscles must be relaxed for palpation.

Generalised abdominal distension is most often explained by the five 'F's:

- Fat
- Fluid (ascites – uncommon in children, most often from nephrotic syndrome)
- Faeces (constipation)
- Flatus (malabsorption, intestinal obstruction)
- Fetus (not to be forgotten after puberty).

Occasionally, it is caused by a grossly enlarged liver and/or spleen or muscle hypotonia.

Causes of localised abdominal distension are:

- Upper abdomen – gastric dilatation from pyloric stenosis, hepato/splenomegaly
- Lower abdomen – distended bladder, masses.

Other signs:

- Dilated veins in liver disease, abdominal striae
- Operative scars (draw a diagram)
- Peristalsis – from pyloric stenosis, intestinal obstruction.

Are the buttocks normally rounded, or wasted as in malabsorption, e.g. coeliac disease or malnutrition?

Palpation

- Use warm hands, explain, relax the child and keep the parent close at hand. First ask if it hurts.
- Palpate in a systematic fashion – liver, spleen, kidneys, bladder, through four abdominal quadrants.
- Ask about tenderness. Watch the child's face for grimacing as you palpate. A young child may become more cooperative if you palpate first with their hand or by putting your hand on top of theirs.

Tenderness

- *Location* – localised in appendicitis, hepatitis, pyelonephritis; generalised in mesenteric adenitis, peritonitis
- *Guarding* – often unimpressive on direct palpation in children. Pain on coughing, on moving about/walking/bumps during car journey suggests peritoneal irritation. Back bent on walking may be from psoas inflammation in appendicitis. By incorporating play into examination, more subtle guarding can be elicited. For example, a child will not be able to jump on the spot if they have localised guarding.

Hepatomegaly (Table 2.4, Fig. 2.8)

- Palpate from right iliac fossa
- Locate edge with tips or side of finger
- Edge may be soft or firm
- Unable to get above it
- Moves with respiration
- Measure (in cm) extension below costal margin in mid-clavicular line.

Percuss downwards from the right lung to exclude pseudohepatomegaly due to lung hyperinflation.

Liver tenderness is likely to be due to inflammation from hepatitis.

Splenomegaly (Table 2.5)

- Palpate from right iliac fossa
- Edge is usually soft
- Unable to get above it
- Notch occasionally palpable if markedly enlarged
- Moves on respiration (ask the child to take a deep breath)
- Measure size below costal margin (in cm) in mid-clavicular line.

If uncertain whether it is palpable:

- Use bimanual approach to spleen
- Turn child onto right side.

A palpable spleen is at least twice its normal size!

Kidneys

These are not usually palpable beyond the neonatal period unless enlarged or the abdominal muscles are hypotonic.

On examination:

- Palpate by balloting bimanually
- They move on respiration
- One can get above them.

Tenderness implies inflammation.

Abnormal masses

- *Wilms' tumour* – renal mass, sometimes visible, does not cross midline
- *Neuroblastoma* – irregular firm mass, may cross midline; the child is usually very unwell
- *Faecal masses* – mobile, non-tender, indentable
- *Intussusception* – acutely unwell, mass may be palpable, most often in right upper quadrant.

Table 2.4 Causes of hepatomegaly

Infection	Congenital, infectious mononucleosis, hepatitis, malaria, parasitic infection
Haematological	Sickle cell anaemia, thalassaemia
Liver disease	Chronic active hepatitis, portal hypertension, polycystic disease
Malignancy	Leukaemia, lymphoma, neuroblastoma, Wilms' tumour, hepatoblastoma
Metabolic	Glycogen and lipid storage disorders, mucopolysaccharidoses
Cardiovascular	Heart failure
Apparent	Chest hyperexpansion from bronchiolitis or asthma

On examining the abdomen:
- **Inspect first, palpate later**
- **Superficial palpation first, deep palpation later**
- **Guarding is unimpressive in children**
- **Silent abdomen – serious!**
- **Immobile abdomen – serious!**

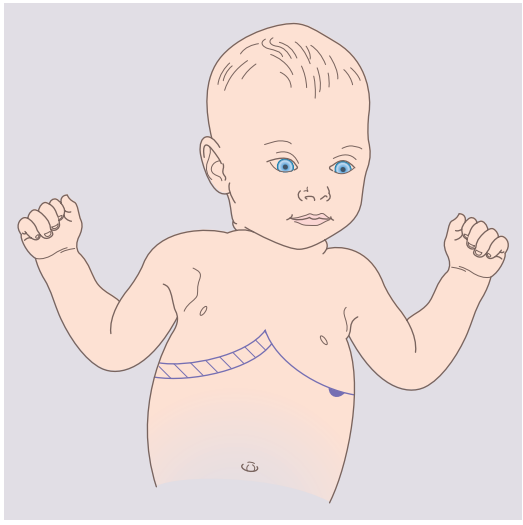


Figure 2.8 Normal findings. The liver edge is 1–2 cm below the costal margin in infants and young children. The spleen may be 1–2 cm below the costal margin in infants.

Table 2.5 Causes of splenomegaly

Infection	Viral, bacterial, protozoal (malaria, leishmaniasis), parasites, infective endocarditis
Haematological	Haemolytic anaemia
Malignancy	Leukaemia, lymphoma
Other	Portal hypertension, systemic juvenile idiopathic arthritis (Still's disease)

Percussion

- *Liver* – dullness delineates upper and lower border. Record span
- *Spleen* – dullness delineates lower border
- *Ascites* – shifting dullness. Percuss from most resonant spot to most dull spot.

Auscultation

Not very useful in 'routine' examination, but important in 'acute abdomen':

- Increased bowel sounds – intestinal obstruction, acute diarrhoea
- Reduced or absent bowel sounds – paralytic ileus, peritonitis.

Genital area

The genital area is examined routinely in young children, but in older children and teenagers this is done only if relevant, e.g. vaginal discharge. Is there an inguinal hernia or a perineal rash?

In *males*:

- Is the penis of normal size?
- Is the scrotum well developed?
- Are the testes palpable? With one hand over the inguinal region, palpate with the other hand. Record if the testis is descended, retractile or impalpable.
- Is there any scrotal swelling (hydrocele or hernia)?

In *females*:

- Do the external genitalia look normal?

Does the anus look normal? Any evidence of a fissure?

Rectal examination

- This should not be performed routinely, and only for specific reasons
- Unpleasant and disliked by children
- Its usefulness in the 'acute abdomen' (e.g. appendicitis) is debatable in children, as they have

a thin abdominal wall and so tenderness and masses can be identified on palpation of the abdomen. Some surgeons advocate it to identify a retrocaecal appendix, but interpretation is problematic as most children will complain of pain from the procedure.
- If intussusception is suspected, the mass may be palpable and stools looking like redcurrant jelly may be revealed on rectal examination.

Urinalysis

- Checked if appropriate
- Clean catch urine specimen preferred and the use of urine bags in the diagnosis of urine tract infection is not advisable
- Dipstick testing for proteinuria, haematuria, glycosuria, leukocyturia
- Examination of the microscopic appearance of urine is helpful for determining the origin of haematuria (crenated red cells, red cell casts).

Lung hyperexpansion in bronchiolitis or asthma may displace the liver and spleen downwards, mimicking hepato/splenomegaly

Neurology/neurodevelopment

Brief neurological screen

A quick neurological and developmental overview should be performed in all children. When doing this:

- Use common sense to avoid unnecessary examination
- Adapt it to the child's age
- Take into consideration the parent's account of developmental milestones.

Watch the child play, draw or write. Are the manipulative skills normal? Can he walk, run, climb, hop, skip, dance? Are the child's language skills and speech satisfactory? Are the social interactions appropriate? Does vision and hearing appear to be normal?

In infants, assess primarily by observation:

- Observe posture and movements of the limbs.
- When picking the infant up, note their tone. The limbs and body may feel normal, floppy or stiff. Head control may be poor, with abnormal head lag on pulling to sitting.

Most children are neurologically intact and do not require formal neurological examination of reflexes, tone, etc. More detailed neurological assessment is performed only if indicated. Specific neurological concerns or problems in development or behaviour require detailed assessment.

More detailed neurological examination

If the child has a neurological problem, a detailed and systematic neurological examination is required.

Patterns of movement

Observe walking and running: normal walking is with a heel–toe gait. Assessment can be incorporated into playing a game, for example: 'pretend you are on a tightrope, how fast can you run?' A toe–heel pattern of walking (toe-walkers) although often idiopathic, may suggest pyramidal tract (corticospinal) dysfunction, a foot drop (common or superficial peroneal nerve lesion), or tight tendo-achilles due to a neuromuscular disease. Children with myopathy may also develop tight Achilles tendon due to weakness. If you are unsure whether a gait is heel–toe or toe–heel, look at the pattern of shoe wear.

A broad-based gait may be due to an immature gait (normal in a toddler) or secondary to a cerebellar disorder. Proximal muscle weakness around the hip girdle can cause a waddling gait. Corticospinal tract lesions give a dynamic pattern of movement involving shoulder adduction, forearm pronation, elbow and wrist flexion with burying of the thumb, whereas internal hip rotation and flexion at the hip and knee and plantar flexion at the ankle give a characteristic circumduction pattern of lower limb movement. If subtle, these are more evident with asking the child to adopt an unusual pattern of walking, e.g. to walk on his heels or toes or with feet inverted. Extrapyramidal lesions give fluctuating tone, with difficulty in initiating or involuntary movements. Look for asymmetry (see Fig. 4.4).

Observe standing from lying down supine. Children up to 3 years of age will turn prone in order to stand because of poor pelvic muscle fixation; beyond this age, it suggests neuromuscular weakness (e.g. Duchenne muscular dystrophy) or low tone, which could be due to a central (brain) cause. The need to turn prone to rise or, later, as weakness progresses, to push off the ground with straightened arms and then climb up the legs is known as Gowers sign (see Fig. 27.6).

Coordination

Assess this by:

- asking the child to build one brick upon another or using a peg-board, and do up and undo buttons, draw, copy patterns, write
- asking the child to hold his arms out straight and close his eyes, and then observing for drift or tremor (this is really looking for asymmetry, position sense, and neglect of one side with visual cues removed)
- finger–nose testing (use teddy's nose to reach out and touch if necessary)
- rapid alternating movements of hands and fingers
- touching tip of each finger in turn with thumb
- asking the child to walk heel–toe, jump and hop.

Subtle asymmetries in gait may be revealed by Fogg's test – children are asked to walk on their heels, the outside and then the inside of their feet. Watch for the pattern of abnormal movement in the upper limbs. Observe them running.

Inspection of limbs

Muscle bulk

- Wasting may be secondary to cerebral palsy, meningomyelocele, muscle disorder or from previous poliomyelitis.
- Increased bulk of calf muscles may indicate Duchenne muscular dystrophy, or myotonic conditions.

Muscle tone

Tone, in limbs

- Best assessed by taking the weight of the whole limb and then bending and extending it around a single joint. Testing is easiest at the knee and ankle joints. Assess the resistance to passive movement as well as the range of movement.
- Increased tone (spasticity) in adductors and internal rotators of the hips, clonus at the ankles or increased tone on pronation of the forearms at rest is usually the result of pyramidal dysfunction. This can be differentiated from the lead-pipe rigidity seen in extrapyramidal conditions, which, if accompanied by a tremor may be termed 'cog-wheel' rigidity.
- The posture of the limbs may give a clue as to the underlying tone, e.g. scissoring of the legs, pronated forearms, fisting, extended legs suggests increased tone (see Figs 4.3 and 4.4). Sitting in a frog-like posture of the legs suggests hypotonia (see Fig. 8.2a), while abnormal posturing and extension suggests fluctuating tone (dystonia).

Truncal tone

- In extra-pyramidal tract disorders, the trunk and head tend to arch backwards (extensor posturing).
- In muscle disease and some central brain disorders, the trunk may be hypotonic (Fig. 2.9a and Fig. 27.8). The child feels floppy to handle and cannot support the trunk in sitting.

Head lag

- This is best tested by pulling the child up by the arms from the supine position (Fig. 2.9b).

Power

Difficult to test in babies. Watch for antigravity movements and note motor function. Both will tell you a lot about power. From 6 months onwards, watch the pattern of mobility and gait. Watch the child standing up from lying and climbing stairs. From the age of 4 years, power can be tested formally against gravity and resistance, first testing proximal muscle and then distal muscle power and comparing sides.

Reflexes

Test with the child in a relaxed position and explain what you are about to do before approaching with a tendon hammer, or demonstrate on parent or toy first. Brisk reflexes may reflect anxiety in the child or a pyramidal disorder. Absent reflexes may be due to a neuromuscular problem or a lesion within the spinal cord, but may also be due to inexpert examination technique. Children will reinforce reflexes if asked.

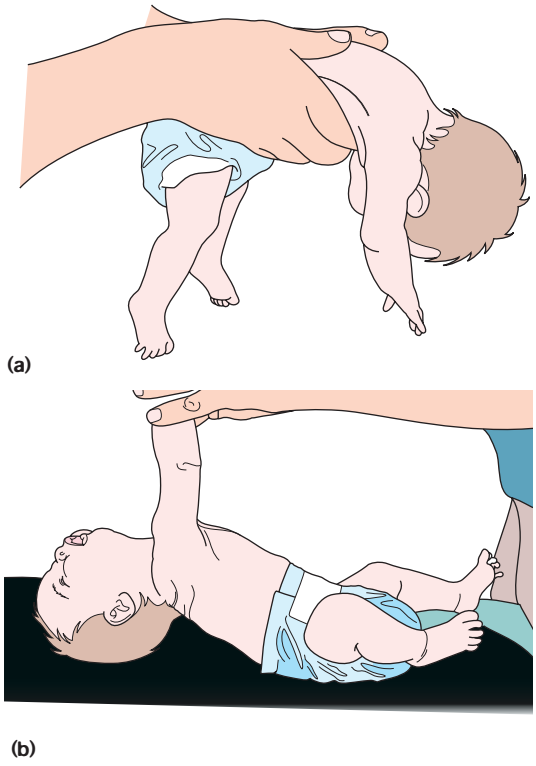


Figure 2.9 A hypotonic infant. **(a)** When held prone, the infant flops like a rag doll. **(b)** Marked head lag on traction of the arms.

Plantar responses

In children the responses are often equivocal and unpopular as it is unpleasant. They are unreliable under 1 year of age. Upgoing plantar responses provide additional evidence of pyramidal dysfunction.

Sensation

Testing the ability to withdraw to tickle is usually adequate as a screening test. If loss of sensation is likely, e.g. meningomyelocele or spinal lesion (transverse myelitis, etc.), more detailed sensory testing is performed as in adults. In spinal and cauda equina lesions there may be a palpable bladder or absent perineal sensation.

Cranial nerves

Before about 4 years old you need some ingenuity to test for abnormal or asymmetric signs – make it a game; ask them to mimic you:

I	Need not be tested in routine practice. Can be done by recognising the smell of a hidden mint sweet.
II	Visual acuity – determined according to age. Direct and consensual pupillary response tested to light and accommodation. Visual fields can be tested if the child is old enough to cooperate.

III, IV, VI	Full eye movement through horizontal and vertical planes. Is there a squint? Nystagmus – avoid extreme lateral gaze, as it can induce nystagmus in normal children.
V	Clench teeth and waggle jaw from side to side against resistance.
VII	Close eyes tight, smile and show teeth.
VIII	Hearing – ask parents, although unilateral deafness could be missed this way. If in doubt, needs formal assessment in a suitable environment.
IX	Levator palati – saying 'aagh'. Look for deviation of uvula.
X	Recurrent laryngeal nerve – listen for hoarseness or stridor.
XI	Trapezius and sternomastoid power – shrug shoulders and turn head against resistance.
XII	Put out tongue and look for any atrophy or deviation.

Bones and joints

A rapid screen to identify disorders of the musculoskeletal is pGALS (paediatric Gait, Arms, Legs, Spine; Fig. 2.10). If an abnormality is found, a more detailed regional examination of the affected joint as well as the joint above and below should be performed (Fig. 2.11).

Neck

Thyroid

- Inspect – swelling uncommon in childhood; occasionally at puberty
- Palpate from behind and front for swelling, nodule, thrill
- Auscultate if enlarged
- Look for signs of hypo/hyperthyroidism.

Lymph nodes

Examine systematically – occipital, cervical, axillary, inguinal. Note size, number, consistency of any glands felt:

- Small, discrete, pea-sized, mobile nodes in the neck, groin and axilla – common in normal children, especially if thin
- Small, multiple nodes in the neck – common after upper respiratory tract infections (viral/bacterial)
- Multiple lymph nodes of variable size in children with extensive atopic eczema – frequent finding, no action required
- Large, hot, tender, sometimes fluctuant node, usually in neck – infected/abscess
- Variable size and shape:
  - infections: viral, e.g. infectious mononucleosis, or TB
  - rare causes: malignant disease (usually non-tender), Kawasaki disease, cat-scratch.

Blood pressure

Indications

Must be closely monitored if critically ill, if there is renal or cardiac disease or diabetes mellitus, or if receiving drug therapy which may cause hypertension, e.g. corticosteroids (Box 2.1). Not measured often enough in children.

Technique

When measured with a sphygmomanometer:

- Show the child that there is a balloon in the cuff and demonstrate how it is blown up.
- Use largest cuff which fits comfortably, covering at least two-thirds of the upper arm.
- The child must be relaxed and not crying.
- Systolic pressure is the easiest to determine in young children and clinically the most useful.
- Diastolic pressure is when the sounds disappear. May not be possible to discern in young children. Systolic pressure used in clinical practice.

Measurement

Must be interpreted according to a centile chart (see Fig. A.3, in the Appendix). Blood pressure is increased by tall stature and obesity. Charts relating blood pressure to height are available and preferable; however, for convenience, charts relating blood pressure to age are often used. An abnormally high reading must be repeated, with the child relaxed, on at least three separate occasions.

Eyes

Examination

Inspect eyes, pupils, iris and sclerae. Are eye movements full and symmetrical? Is nystagmus detectable? If so, may have ocular or cerebellar cause, or testing may be too lateral to the child. Are the pupils round (absence of posterior synechiae), equal, central and reactive to light? Is there a squint? (see Ch. 4).

Epicanthic folds are common in Asian ethnic groups.

Ophthalmoscopy

- In infants, the red reflex is seen from a distance of 20–30 cm. Absence of red reflex occurs in corneal clouding, cataract, retinoblastoma.
- Fundoscopy – difficult. Requires experience and cooperation. In infants, mydriatics are needed and an ophthalmological opinion may be required. Retinopathy of prematurity and retinopathy of

Box 2.1 Measuring blood pressure in children

- Sphygmomanometer (Fig. 2.12)
  - stethoscope in older children
  - Doppler ultrasound in infants
- Oscillometric (e.g. Dynamapp) – unreliable in infants and young children
- Invasive – direct measurement from an arterial catheter is preferable if critically ill.

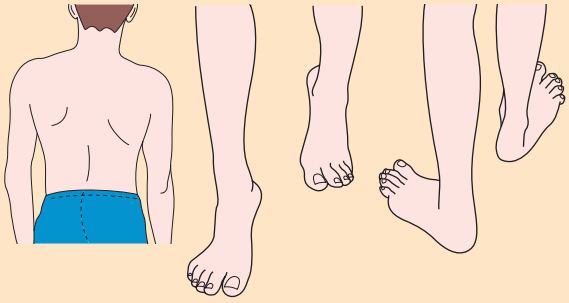
pGALS – musculoskeletal screening for school-aged children

(Differences from adult GALS highlighted in bold)

Screening questions

- **Do you (or your child) have any pain or stiffness in your joints, muscles or your back?**
- **Do you (or your child) have any difficulty getting yourself dressed without any help?**
- **Do you (or your child) have any difficulty going up and down stairs?**

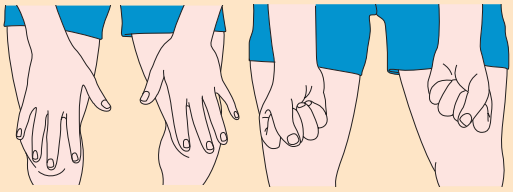
POSTURE AND GAIT



Observe standing (from front, back and sides)

Observe walking **'Walk on your tip-toes, walk on your heels'**

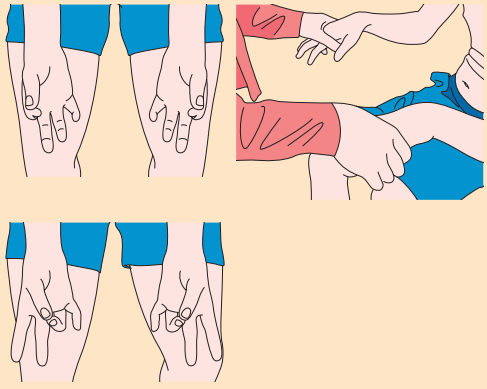
ARMS



'Put your hands out straight in front of you'

'Turn your hands over and make a fist'

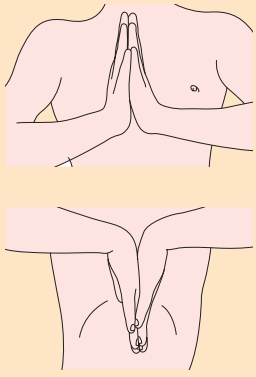
ARMS



'Pinch your index finger and thumb together'

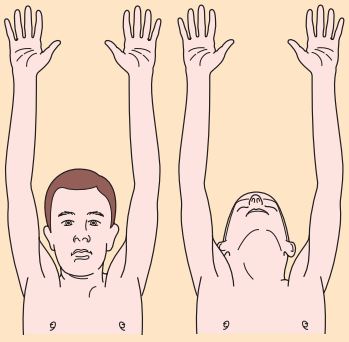
'Touch the tips of your fingers with your thumb'

Squeeze the metacarpophalangeal joints for tenderness



'Put your hands together palm to palm'

'Put your hands back to back'



'Reach up and touch the sky'

'Look at the ceiling'

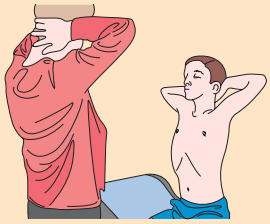
Figure 2.10 pGALS (paediatric Gait, Arms, Legs, Spine) musculoskeletal screening for school-aged children (Foster HE et al., Musculoskeletal screening examination (pGALS) for school-aged children based on the adult GALS screen. Arthritis Rheum 2006; 55(5):709-16 and see http://www.arthritisresearchuk.org/videos/pgals.aspx to view video of the examination)

Continued

pGALS – musculoskeletal screening for school-aged children—cont'd

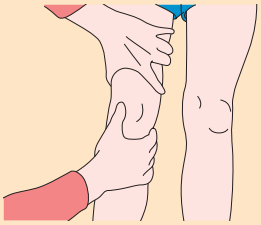
(Differences from adult GALS highlighted in bold)

ARMS



'Put your hands behind your neck'

LEGS



Feel for effusion at the knee



'Bend and then straighten your knee' (Active movement of knees and feel for crepitus)



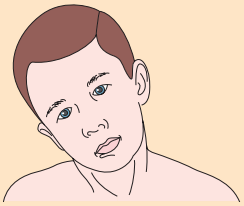
Passive movement of hip

TEMPOROMANDIBULAR JOINT



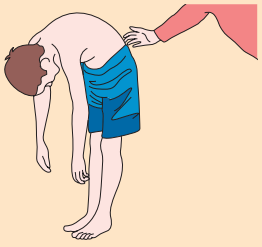
'Open your mouth and put three fingers in your mouth'

NECK AND SPINE



'Touch your shoulder with your ear'

Observe lateral flexion of cervical spine



'Bend forward and touch your toes'

Observe curve of the spine

Figure 2.10, cont'd

Regional musculoskeletal assessment

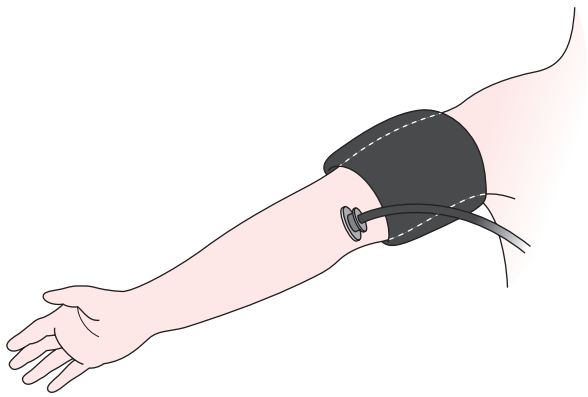
Look:
- For signs of discomfort
- Skin abnormalities – rashes, scars, bruising, colour, nail abnormalities
- Limb alignment, leg length, muscle bulk and evidence of asymmetry
- Bony deformity, soft tissue, joint swelling or muscle changes

Feel:
- Each joint, long bones and neighbouring soft tissues:
- Palpate along bones and joint line for tenderness
- Feel for warmth ***(infection or inflammation)***
- Delineate bony or soft tissue swellings
- Check for joint effusion, most readily at the knee

Move:
- For each joint, ask to move the joint first (active movement).
  Observe for discomfort, symmetry and range of movement.
- Passively move the joint, noting range of any restriction of movement (compare sides but note bilateral changes)
- Lateral and rotational movements may be as important as flexion and extension.

Function:
- For lower limb joints – check gait
- For small joints such as hands - check grip

Figure 2.11 A regional musculoskeletal assessment. (See Foster HE and Brogan P. *Oxford Handbook of Paediatric Rheumatology,* Oxford, 2011, Oxford University Press and http://www.arc.org.uk/arthinfo/medpubs/6321/6321.asp)



Cuff >2/3 upper arm.
(Smaller cuffs give artificially high readings)

Age	Upper limit of normal systolic blood pressure
1–5 years	110 mmHg
6–10 years	120 mmHg

Figure 2.12 Measurement of blood pressure.

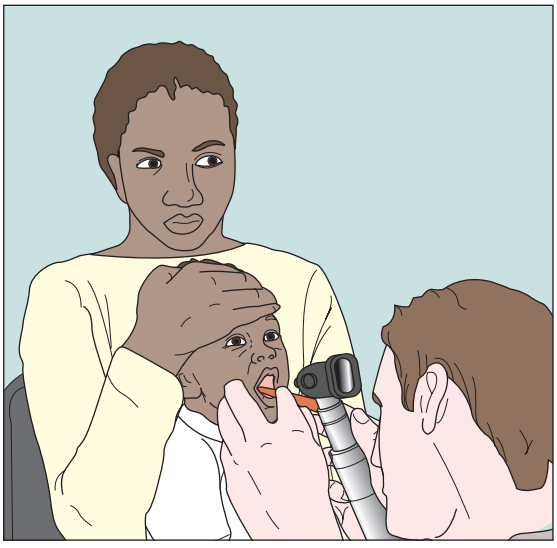


Figure 2.13 Holding a young child to examine the throat. The mother has one hand on the head and the other across the child's arms.



Figure 2.14 Holding a young child correctly is essential for successful examination of the ear with an auroscope. The mother has one hand on the child's head and the other hand holding the upper arm.

congenital infections and choroido-retinal degeneration show characteristic findings. Retinal haemorrhages may be seen in head trauma or in 'shaken baby syndrome' (non-accidental injury).

- In older children with headaches, diabetes mellitus or hypertension, optic fundi should be examined. Mydriatics are not usually needed.

Ears and throat

Examination is usually left until last, as it can be unpleasant. Explain what you are going to do. Show the parent how to hold and gently restrain a younger child to ensure success and avoid possible injury (Figs 2.13 and 2.14).

Throat

Try quickly to get a look at the tonsils, uvula, pharynx and posterior palate. Older children (5 years +) will open their mouths as wide as possible without a spatula. A spatula is required for young children. Look for redness, swelling, pus or palatal petechiae. Also check the teeth for dental caries and other gross abnormalities.

Ears

Examine ear canals and drums gently, trying not to hurt the child. Look for anatomical landmarks on the ear drum and for swelling, redness, perforation, dullness, fluid.

Communicating with children

Throughout the consultation, make sure that your communication with the child is appropriate for the child's age and stage of development (Table 2.6).

Table 2.6 The reasons for talking with children

Why talk to children when you can get the information from the parent? The reasons are:

- To establish rapport
- To obtain the child's own views about their problems
- To know how the child feels about their health and life
- To reduce anxiety and fear and to improve compliance with assessment and treatment
- To determine the presence of associated emotional or psychiatric problems

	Preschool child (2–5 years)	**School-age child (6–11 years)**	**Adolescent (12–18 years)**
Thought processes	When I close my eyes, Mum goes away (world viewed differently, from own perspective) I am asleep, so everyone is asleep (centre of their world) When I fell, the floor hurt me (objects are alive) My toy elephant is crying because the other elephants won't play with him (involvement in pretend play)	I have been invited to Katie and Jane's parties - maybe I could go for some time to each (able to start solving concrete problems) Am I going to be chosen for the school choir? (develops worries about the future) Mum gets really upset when Dad gets drunk, but Dad does not care (able to see another person's point of view and take on more than one perspective)	I can handle things without Mum's help (seeking autonomy and separation) Should our country be at war? (develops concern about social issues)
Effect on the way we talk to them	Use short, concrete questions within their immediate experience. To avoid yes/no answers use a choice of options, e.g. when you go to nursery, what do you like to do – draw or dress up or something else? Use toys or puppets while interviewing, e.g. to represent different people in the child's life	Use familiar examples of experience of others to explore the child's feelings and behaviour, e.g. when a boy was bullying another boy at school, he came to see me so we could talk about how he controls his temper. Do you ever get angry and bully others? You can get at their hopes and dreams by asking them, 'If I was a magician and could give you three wishes, what would they be?'	Should be given an opportunity to be seen alone as they may have problems and difficulties not known to the parents and that the adolescent does not want to share with them Upsetting thoughts can be explored in some adolescents using metaphors







Summary

In taking a history and performing a clinical examination:

- The child's age is a key feature – it will determine the nature of the problem, how the consultation is conducted, the likely diagnosis and its management.
- The interview environment should be welcoming – with suitable toys for young children.
- Most information is usually obtained from a focused history and observation, rather than detailed examination although examination is also important.
- Check growth, including charts in personal child health record, and development.
- With young children – be confident but gentle, do not ask their permission to examine them or they may say 'no', and leave unpleasant procedures (ears and throat) until last.
- Involve children with the consultation, as appropriate to their age.

Remember child protection when taking a history or examining a child where there are unusual findings.

Summary and management plan

At the end of the history and examination:

- Summarise the key problems (in physical, emotional, social and family terms, if relevant).
- List the diagnoses or differential diagnoses. Draw up a management plan to address the problems, both short and long term. This could be reassurance, a period of observation, performing investigations or therapeutic intervention.
- Provide explanation to the parents and to the child, if old enough. Consider providing further information, either written or on the internet.
- If relevant, discuss what to tell other members of the family.
- Consider which other professionals should be informed.
- Write a brief summary in the child's personal child health record.
- Ensure your notes are dated and signed.

Further reading

Gill D, **O'Brien N:** *Paediatric Clinical Examination Made Easy*, ed 5, Edinburgh, 2007, Churchill Livingstone.

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3

Normal child development, hearing and vision

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Children acquire functional skills throughout childhood. The term 'child development' is used to describe the skills acquired by children between birth and about 5 years of age, during which there are rapid gains in mobility, speech and language, communication and independence skills. During school age, evidence of developmental progression is predominantly through cognitive development and abstract thinking, although there is also some further maturation of early developmental skills.

Normal development in the first few years of life is monitored:

- by parents, who are provided with guidance about normal development in their child's personal child health record and in a book *Birth to Five*, given to all parents in the UK
- at regular child health surveillance checks
- whenever a young child is seen by a healthcare professional, when a brief opportunistic overview is made.

The main objective of assessing a young child's development is the early detection of delayed or abnormal development in order to:

- help children achieve their maximum potential
- provide treatment or therapy promptly (particularly important for impairment of hearing and vision)
- act as an entry point for the care and management of the child with special needs.

This chapter covers normal development. Delayed or abnormal development and the child with special needs are considered in Chapter 4.

Influence of heredity and environment

A child's development represents the interaction of heredity and the environment on the developing brain. Heredity determines the potential of the child, while the environment influences the extent to which that potential is achieved. For optimal development, the environment has to meet the child's physical and psychological needs (Fig. 3.1). These vary with age and stage of development:

- Infants are totally physically dependent on their parents and require a limited number of carers to meet their psychological needs.
- Primary school-age children can meet some of their physical needs and cope with many social relationships.
- Adolescents are able to meet most of their physical needs while experiencing increasingly complex emotional needs.

Fields of development

There are four fields of developmental skills to consider whenever a young child is seen (Fig. 3.2):

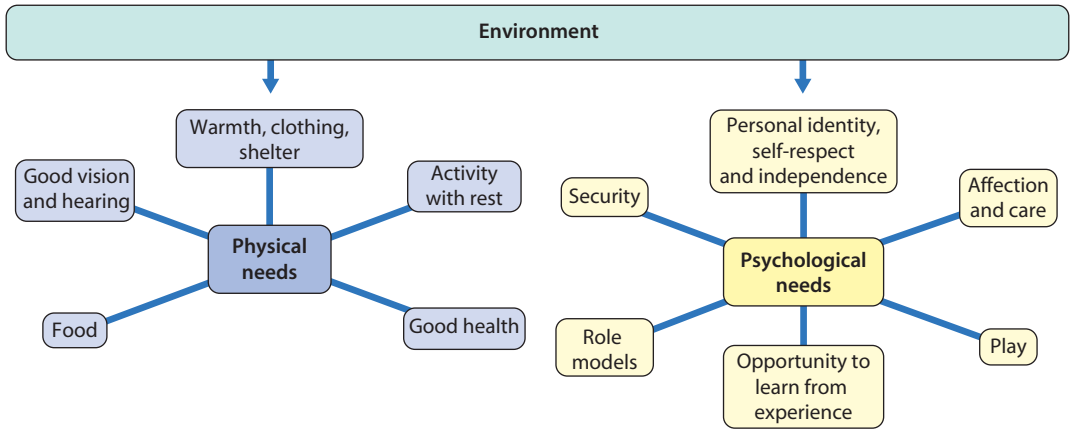


Figure 3.1 Development can be impaired if the environment fails to meet the child's physical or psychological needs.

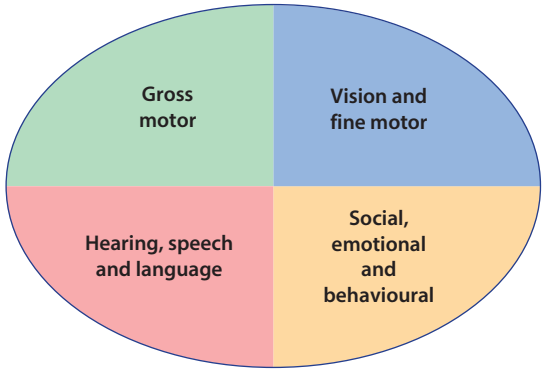


Figure 3.2 The four functional areas of child development and their core features.

- Gross motor
- Vision and fine motor
- Hearing, speech and language
- Social, emotional and behavioural.

Gross motor skills are the most obvious initial area of developmental progress. As fine motor skills require good vision, these are grouped together; similarly, normal speech and language development depends on reasonable hearing and so these are also considered together. Social, emotional and behavioural skills are a spectrum of psychological development.

The acquisition of developmental abilities for each skill field follows a remarkably constant pattern between children, but may vary in rate. It is like a sequential story. Thus, the normal pattern for acquisition of skills:

- is sequentially constant
- should always be considered longitudinally, relating each stage to what has gone before and what lies ahead
- varies in rate between children.

A deficiency in any one skill area can have an impact on other areas. For instance, a hearing impairment may affect a child's language, social and communication skills and behaviour. As a child grows, additional skills become important, such as attention and concentration and how an individual child manages to integrate their skills.

Developmental milestones

Chronological age, physical growth and developmental skills usually evolve hand in hand. Just as there are normal ranges for changes in body size with age, so there are ranges over which new skills are acquired. Important developmental stages are called developmental milestones.

When considering developmental milestones:

- The *median age* is the age when half of a standard population of children achieve that level; it serves as a guide to when stages of development are likely to be reached but does not tell us if the child's skills are outside the normal range.
- *Limit ages* are the age by which they should have been achieved. Limit ages are usually 2 standard deviations (SD) from the mean. They are more useful as a guide to whether a child's development is normal than the median ages. Failure to meet them gives guidance for action regarding more detailed assessment, investigation or intervention.

Median and limit ages

The difference between median and limit ages is shown by considering the age range for the developmental milestone of walking unsupported. The percentage of children who take their first steps unsupported is:

- 25% by 11 months
- 50% by 12 months
- 75% by 13 months
- 90% by 15 months
- 97.5% by 18 months.

The median age is 12 months and is a guide to the common pattern to expect, although the age range is wide. The limit age is 18 months (2 SDs from the mean).

Of those not achieving the limit age, many will be normal late walkers, but a proportion will have an underlying problem, such as cerebral palsy, a primary muscle disorder or global developmental delay. A few may be understimulated from social deprivation. Hence, any child who is not walking by 18 months should be assessed and examined. Thus, 18 months can be set as a 'limit age' for children not walking. Setting the limit age earlier may allow earlier identification of problems, but will also increase the number of children labelled as 'delayed' who are in fact normal.

Variation in the pattern of development

There is variation in the pattern of development between children. Taking motor development as an example, normal motor development is the progression from immobility to walking, but not all children do so in the same way. While most achieve mobility by crawling (83%), some bottom-shuffle and others crawl with their abdomen on the floor, so-called commando crawling (creeping) (Fig. 3.3). A very few just stand up and walk. The locomotor pattern (crawling, creeping, shuffling, just standing up) determines the age of sitting, standing and walking.

The limit age of 18 months for walking applies predominantly to children who have had crawling as their early mobility pattern. Children who bottom-shuffle or commando crawl tend to walk later than crawlers, so that within those not walking at 18 months there will be some children who demonstrate a locomotor variant pattern, with their developmental progress still being normal. For example, of children who become mobile by bottom-shuffling, 50% will walk independently by 18 months and 97.5% by 27 months of age, with even later ages for those who initially commando crawl.

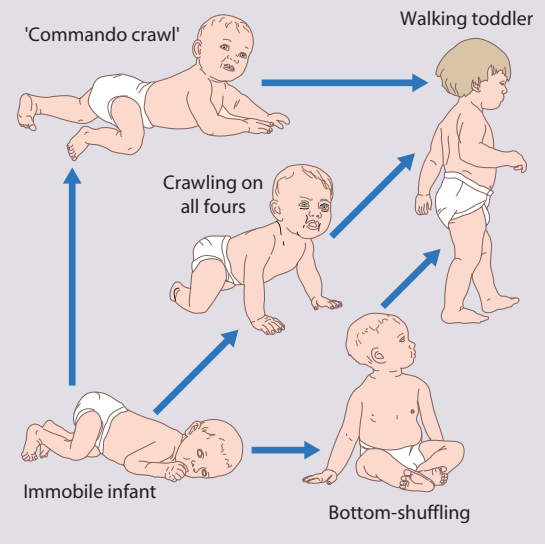


Figure 3.3 Early locomotor patterns. Most children crawl on all fours prior to walking, but some 'bottom-shuffle' and others 'commando crawl' (creep). Bottom-shuffling often runs in families. The late walking that often goes with this locomotor variant needs to be differentiated from an abnormality such as cerebral palsy.

Adjusting for prematurity

If a child has been born preterm, this should be allowed for when assessing developmental age by calculating it from the expected date of delivery. Thus the anticipated developmental skills of a 9-month-old baby (chronological age) born 3 months early at 28 weeks' gestation are more like those of a 6-month-old baby (corrected age). Correction is not required after about 2 years of age when the number of weeks early the child was born no longer represents a significant proportion of the child's life.

Is development normal?

When evaluating a child's developmental progress and considering whether it is normal or not:

- Concentrate on each field of development (gross motor; vision and fine motor; hearing, speech and language; social, emotional and behavioural) separately.
- Consider the developmental pattern by thinking longitudinally and separately about each

Summary

Assessing child development

When assessing a young child's development:

- Consider the four fields of developmental skills: gross motor; vision and fine motor; hearing, speech and language; social, emotional, behavioural
- The acquisition of developmental abilities follows a similar pattern between children, but may vary in rate, and still be normal.

Terms used are:

- Developmental milestones: the acquisition of important developmental skills
- Median age – when half the population acquire a skill; serves as a guide to normal pattern of development
- Limit age – when a skill should have been acquired; further assessment is indicated if not achieved.

When evaluating a child's development, consider:

- the sequence of developmental progress
- the stage the child has reached for each skill field
- if progress is similar in each skill field
- how the child's developmental achievements relate to age.

developmental field. Ask about the sequence of skills achieved as well as those skills to be anticipated shortly.
- Determine the level the child has reached for each skill field.
- Now relate the progress of each developmental field to the others. Is the child progressing at a similar rate through each skill field, or does one or more field of development lag behind the others?
- Then relate the child's developmental achievements to age (chronological or corrected).

This will enable you to decide if the child's developmental progress is normal or delayed. Normal development implies steady progress in all four developmental fields with acquisition of skills occurring before limit ages are reached. If there is developmental delay, does it affect all four developmental fields (global delay), or one or more developmental field only (specific developmental delay)? As children grow older and acquire further skills, it becomes easier to make a more accurate assessment of their abilities and developmental status.

Pattern of child development

This is shown pictorially for each field of development, including key developmental milestones and limit ages:

- Gross motor development (Fig. 3.4 and Table 3.1)
- Vision and fine motor (Fig. 3.5)
- Hearing, speech and language (Fig. 3.6)
- Social, emotional and behavioural (Fig. 3.7).

In order to screen a young child's development, it is necessary to know only a limited number of key developmental milestones and their limit ages.

Cognitive development

Cognition refers to higher mental function. This evolves with age. In infancy, thought processes are centred around immediate experiences. The thought processes of preschool children (which have been called pre-operational thought by Piaget, who described children's intellectual development) tend to be that:

- they are the centre of the world
- inanimate objects are alive and have feelings and motives
- events have a magical element
- everything has a purpose. Toys and other objects are used in imaginative play as aids to thought, to help make sense of experience and social relationships.

In middle-school children, the dominant mode of thought is practical and orderly, tied to immediate circumstances and specific experiences. This has been called operational thought.

It is only in the mid-teens that an adult style of abstract thought (formal operational thought) begins to develop, with the ability for abstract reasoning, testing hypotheses and manipulating abstract concepts.

Table 3.1 The primitive reflexes present at birth gradually disappear as postural reflexes develop, which are essential for independent sitting and walking

Primitive reflexes	Postural reflexes
Moro – sudden extension of the head causes symmetrical extension, then flexion of the arms	**Labyrinthine righting** – head moves in opposite direction to which the body is tilted
Grasp – flexion of fingers when an object is placed in the palm	**Postural support** – when held upright, legs take weight and may push up (bounce)
Rooting – head turns to the stimulus when touched near the mouth	**Lateral propping** – in sitting, the arm extends on the side to which the child falls as a saving mechanism
Stepping response – stepping movements when held vertically and dorsum of feet touch a surface	**Parachute** – when suspended face down, the arms extend as though to save themself
Asymmetrical tonic neck reflex – lying supine, the infant adopts an outstretched arm to the side to which the head is turned	

Analysing developmental progress

Detailed assessment

So far, emphasis has been mainly on thinking about developmental progress in a longitudinal way, taking each skill field and its progression separately, and then relating the progress in each to the others and to chronological age. This is the fundamental concept of learning how to think about developmental assessment of children. Detailed questioning and observation is required to assess children with developmental problems but is unnecessary when screening developmental progress in normal clinical practice, when a short-cut approach can be adopted.

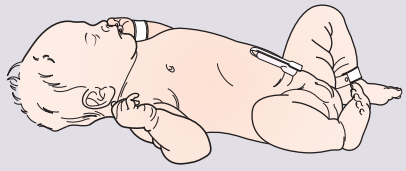
The short-cut approach

This concentrates on the most actively changing skills for the child's age. The age at which developmental progress accelerates differs in each of the developmental fields. Figure 3.8 demonstrates the age when there is the most rapid emergence of skills in each developmental field. This means there is for:

- gross motor development: an explosion of skills during the first year of life

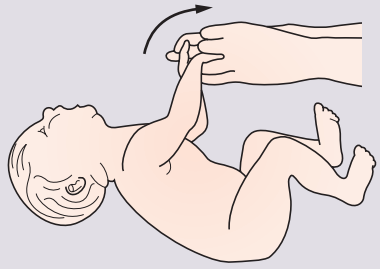
Gross motor development (median ages)

newborn



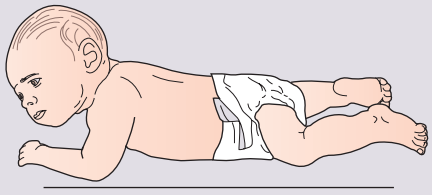
Limbs flexed, symmetrical posture

newborn



Marked head lag on pulling up

6–8 weeks



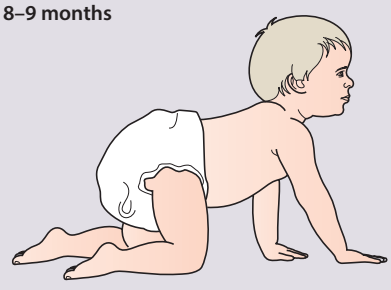
Raises head to 45° in prone

6–8 months



Sits without support
– at 6 months: with round back
– at 8 months: with straight back (shown)

8–9 months



Crawling

10 months



Cruises around furniture

12 months



Walks unsteadily,
broad gait, hands apart

15 months



Walks steadily

Figure 3.4 Gross motor development (median ages).